A Computationally Feasible Framework for Causal Probabilistic Explanation

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Abstract

Explaining why a specific outcome occurred, and which inputs deserve the blame or credit, is central to philosophical, scientific, and policy analysis. Existing tools split into two camps. The Halpern–Pearl theory of *actual causality* gives principled verdicts, but only for toy-sized models, because computing them requires enumerating counterfactual scenarios. Scalable attribution methods like SHAP and LIME ignore the causal structure that generated the data, and can give answers that conflict with a careful causal analysis. We close this gap with Probabilistic Causal Impact (PCI).

PCI builds on actual causality and on Pearl's notions of probability of necessity and sufficiency, but recasts the question "which variables explain this outcome?" as an estimation problem on a probabilistic causal model that one can solve by approximate methods, such as MC. To compute a PCI score, the analyst specifies three ingredients: a distribution Γ over which subsets of variables to consider as candidate explanations, a distribution Δ over the counterfactual values to which those variables might be changed, and a function ci that measures how much such a change shifts the outcome. The result is a tractable, causally grounded explanation procedure that applies to large models. A correspondence theorem shows that PCI agrees with actual causality on the small models where AC is well-defined; against Pearl's probability of actual causation, PCI recovers Pearl's within-scenario ranking and additionally captures path-reliability and cross-scenario reliability, intuitions Pearl's binary indicator cannot capture.

We evaluate PCI in four increasingly demanding settings: a synthetic structural model with redundant and preempting causes, where PCI recovers the AC literature's verdicts and generalises them to continuous variables; a scaling benchmark where PCI remains tractable as AC enumeration becomes infeasible; a dynamical (Bayesian, ODE) epidemiological model with continuous outcomes and asymmetric causes, where AC offers no verdict; and a deployed spatio-temporal model with deep ML components trained on millions of points, where PCI and SHAP attributions diverge in the same qualitative ways as in the synthetic setting.

Keywords: Probabilistic Causal Models, Explanation, Responsibility, Actual Causality, Probability of Necessity and Sufficiency.

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1 Introduction

Causal attribution is a central reasoning task in many important problems, such as gauging the efficacy of medical treatments [Shepherd et al., 2024], pinpointing drivers of economic growth [Cheshire and Magrini, 2009], root cause analysis in automated systems [Papageorgiou et al., 2022], or assessing the impact of public policy [Heckman and Pinto, 2022]. We may want to understand why an adverse outcome occurred so as to prevent recurrence, or how a predicted positive outcome could be brought about.

In the absence of scalable, formal causal attribution tools, analysts often resort to ad-hoc methods: strong correlation, historical comparison, domain expertise, or non-causal attribution like Granger causality [Granger, 1969] and SHAP [Lundberg and Lee, 2017]. Even researchers with an advanced command of statistical and experimental methods may not possess a thorough understanding of how to ascribe cause [Shepherd et al., 2024], and would benefit from scalable automation. Here we present a tool for accurate and tractable causal attribution applicable to a wide array of causal, probabilistic, machine-learned models.

What would a satisfactory causal attribution method look like? At minimum, it should be (i) *causally grounded*, tracking counterfactual dependence rather than correlation; (ii) *context-sensitive*, distinguishing causes whose path was active in the factual world from those whose path was blocked; (iii) *broadly applicable*, handling stochastic models, discrete and continuous variables, and finite (not just infinitesimal) counterfactual changes; and (iv) *tractable*, estimable from samples without combinatorial enumeration over witness sets and alternatives. Existing methods each forfeit one or more of these. Actual causality [Halpern, 2016] is grounded and context-sensitive, but neither broadly applicable nor tractable in general, with no clear approximation [Eiter and Lukasiewicz, 2002]. Methods that *do* scale either give up causal grounding or are restricted in scope. SHAP [Lundberg and Lee, 2017] grounds attribution in observational correlations, so a feature can receive high attribution merely for being correlated with a true cause; Causal SHAP [Heskes et al., 2020] addresses this to some extent, but failure modes persist (Section 4). Butler et al. [2022]'s Differential Causal Effect (DCE) is causally grounded but local: it relies on gradients, so requires differentiability and does not capture effects from non-infinitesimal changes in potential causes.[6] Standard treatment-effect quantities, ATE and CATE, are causally grounded and tractable but lack context-sensitivity: every intervention propagates downstream through the structural equations, overriding the mediator values that mark which causal paths were active in the factual world (Section 2).

We introduce *probabilistic causal impacts* (PCI), an attribution framework that combines the context-sensitivity of actual causality with the scalability of probability-of-causation methods. PCI generalises Pearl's probability of necessary and sufficient causation [Pearl, 2022] from binary to arbitrary variable domains, and incorporates the *witness* mechanism from actual causality [Halpern, 2016] via integration over a distribution of witness sets rather than existential quantification.

PCI also shares an intuition with the literature on *descriptive normality* [Halpern and Hitchcock, 2015, Icard et al., 2017]: attributions are computed as expectations against a distribution over counterfactual worlds conditioned on upstream factual context, so factual upstream values play the role of a default against which alternatives are graded. We do not aim to resolve the specific counterexamples that motivate the normality literature; the connection is at the level of why a non-uniform alternative-value distribution matters, not in handling explicit normality orderings. Several components have precedents — expectations over exogenous noise [Halpern, 2016], weighting by alternative-value probabilities [Beckers and Vennekens, 2015], aggregation over witness sets [Beckers and Vennekens, 2015, Halpern, 2016] — but no prior construction combines them into a single tractable estimator handling arbitrary variable domains.

PCI presumes access to a trained probabilistic causal model from which counterfactual outcomes can be sampled, a stronger requirement than methods that work from observational data and a causal graph alone (e.g., PN/PS/PNS bounds, Causal SHAP). Within that requirement, however, we place no strong restrictions on the form of the model, distributions, or types of variables. Such models can comprise standard machine learning components such as neural network parametrized distributions or Gaussian processes.

[6] DCE is, fairly, not designed for responsibility attribution: Butler et al. target rates of change of treatment.

With this breadth of application, and a measure that surfaces its design choices — the variable selection distribution, alternative value distribution, and causal impact function — as explicit user-facing components rather than hidden defaults, PCI hopes to provide causal attribution and explanation tooling for a wide array of users and applications.

Paper structure. The paper proceeds in four phases. The first establishes motivation and formalism. Section 2 works through a running example to surface what actual causality, the probability of necessity/sufficiency, and average/conditional treatment effects each get right and miss, motivating the design choices behind PCI. Section 3 then defines the method: a kernel built from a sufficiency check at the factual configuration and a necessity check against alternatives, integrated against a distribution over suspect/witness sets, with three components exposed as user choices: the variable selection distribution Γ, the alternative-value distribution Δ, and the causal impact function.

The second phase positions PCI against existing attribution methods. Section 4 compares PCI with SHAP, Causal SHAP, ATE, and CATE on two small examples (OBCB, with two binary features, and a signal-with-mediation model), tabulating which desiderata each satisfies or fails, including cases where SHAP and Causal SHAP attributions track correlation rather than counterfactual dependence. Section 5 reports a synthetic evaluation built around overdetermination and undercutting, the patterns on which pre-AC accounts most often diverge from intuition, and shows that PCI recovers the literature's verdicts. Section 6 contrasts PCI with gradient-based attribution, instantiated by Differential Causal Effect [Butler et al., 2022], on a continuous credit-limit example, showing where the two diverge and why PCI better tracks responsibility under unit rescaling and outside the steep region of the response surface.

The third phase establishes the formal and empirical relationship between PCI and actual causality. Section 7 shows that under specific choices of Γ and Δ the PCI kernel is positive on a configuration exactly when that configuration is an actual cause in Halpern's sense, so AC verdicts can be read off the kernel without enumerating witnesses. Section 8 positions PCI against Pearl's probability of actual causation: on the canonical desert-traveller example, PCI recovers Pearl's within-scenario ranking with graded magnitudes, and a weak-poison variant surfaces two additional responsibility intuitions (path-reliability and cross-scenario reliability) that Pearl's binary indicator structurally cannot express but PCI's expectation does. Section 9 then benchmarks PCI against explicit actual-cause computation: exact subset enumeration times out at 17 variables, while the necessity-only PCI estimator continues to roughly 73–145 variables (depending on sample budget) and sustains above a 0.85 correct-attribution rate up to about 60 variables, visiting several orders of magnitude fewer cause–witness configurations.

The fourth phase pushes PCI into settings where actual causality no longer applies. Section 10 extends the benchmarking to a continuous-outcome dynamical-SIR (Susceptible–Infected–Recovered) setting, exercising the joint necessity-sufficiency machinery on threshold events (peak overshoot) where AC has no comparable verdict. Where but-for analysis cannot separate the two policies, lockdown and masking, PCI correctly assigns lockdown roughly twice the responsibility of masking, with the gap driven entirely by the necessity term under witness pinning. Section 11 applies PCI to a deployed automated valuation model with machine-learned components trained on millions of points, a regime where actual causality is intractable; SHAP returns qualitatively different attributions, assigning virtually all responsibility to downstream variables and effectively ignoring the rest. Section 12 closes with discussion.

2 Causal Impact: Motivations

The central difficulty in causal attribution is not computing counterfactuals — it is knowing which counterfactuals to compute, and recognising when a cause is blocked by context rather than absent altogether. The closest predecessors of this approach, taken individually, fail to capture this: actual causality handles context sensitivity but is computationally intractable and not general enough; the probability of necessity and sufficiency scales well, but lacks context sensitivity. This section builds intuition for both limitations through a running example, setting up the design choices that motivate PCI in Section 3. We proceed in three steps: revisit Halpern's actual-cause construction on a deterministic OBCB example to see what witnesses do and why a plain but-for clause misses Alice's case; turn to Pearl's PN/PS/PNS on a stochastic OBCB to introduce the probabilistic machinery actual causality lacks while showing that, without witnesses, PN/PS/PNS cannot register Alice's missing credit check; and close with a "Path forward" subsection that distils what PCI must inherit from each predecessor before Section 3 formalises it.

2.1 Actual Causality

Consider the following (deterministic, discrete) model:

Example 1 (Bob at Old Boys' Club Bank)**.** *Bob applied for a home mortgage loan with the local branch of Old Boys' Club Bank (OBCB), and his application was denied. The OBCB policy requires applicants to pass a credit score check, but it only performs this check if the applicant is male; all non-male applicants are rejected outright. Thus, the procedure is:*

$$\begin{aligned} \mathit{check} &= (\mathit{gender} = \mathit{male}) \\ \mathit{check\text{-}failed} &= \mathit{check} \wedge (\mathit{credit} = \mathit{bad}) \\ \mathit{loan} &= \neg \mathit{check\text{-}failed} \wedge (\mathit{gender} = \mathit{male}) \end{aligned}$$

Why was Bob's loan application denied? An intuitive explanation is that his credit was bad: had it been good, the outcome would have been different. In contrast, his gender does not appear to be responsible for the rejection: had it been different, his application still would have been denied (albeit this time due to $\mathit{check} = F$).

This suggests a counterfactual approach to feature responsibility attribution: perhaps we can identify the features responsible for a particular outcome by asking counterfactual questions: would the outcome have been different if a specific feature had a different value? This is called the *but-for clause*: Bob wouldn't have been denied the loan but for his bad credit.

Concretely, consider a model M with *exogenous variables* $\mathbf{U}$, which are the source of stochasticity, and *endogenous variables* $\mathbf{V}$, which are influenced by other variables in the model. Let $\mathbf{S} \subseteq \mathbf{V}$ be the variables we suspect to be causally responsible (call them *causal suspects*), and $Y \in \mathbf{V}$ be an outcome of interest. Suppose we consider only one suspect $S \in \mathbf{S}$ and mark factual values with *. Under a setting $\mathbf{U} = \mathbf{u}$ of the exogenous variables, M produces $S = s^\star$ and $Y = y^\star$, denoted $(M, \mathbf{u}) \models S = s^\star, Y = y^\star$. We call this the *factual* scenario. An intervened model where a variable S is set to an alternative value s' is denoted $(M, \mathbf{u})[S \leftarrow s']$.

A first attempt at defining a counterfactual notion of causal responsibility is:

Definition 2 (But-for, single)**.** *$S = s^\star$ is causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models S = s^\star, Y = y^\star$ and $(M, \mathbf{u})[S \leftarrow s'] \not\models Y = y^\star$ for some $s' \neq s^\star$.*

In a given context, S is causally responsible for an outcome only when intervening on S with some alternative value would change that outcome. Bob's credit score satisfies this condition: had his credit score been higher, the loan might have been approved.

However, this cause test is too coarse. Consider another case:

Example 3 (Alice at OBCB)**.** *Alice is a woman with bad credit. She applies for a loan with OBCB and is rejected.*

Why was Alice denied the loan? Under the but-for clause for single cause candidates, neither her gender nor her credit score are responsible. No matter how high her credit score, her application would still have been denied because the bank does not loan to women. Had her gender been different, she would still have been denied for having bad credit.

To address this example, we could consider sets of factors as causes:

Attempt 4 (But-for, sets)**.** *Variables $\mathbf{C} \subseteq \mathbf{S}$ with factual values $\mathbf{C} = \mathbf{c}^\star$ are causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^\star, Y = y^\star$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}'] \not\models Y = y^\star$ for some $\mathbf{c}'$ everywhere different from $\mathbf{c}^\star$, where no proper subset of $\mathbf{C}$ has this property.*

The minimality condition reflects our preference for explanations that include only relevant properties.

Under this generalization, the set $\mathbf{c}^\star = \{\mathit{gender} = \text{female}, \mathit{credit} = \text{bad}\}$ is jointly responsible for Alice's application being denied because there exists $\mathbf{c}' = \{\mathit{gender} = \text{male}, \mathit{credit} = \text{good}\}$ that would change the decision and $\mathbf{c}^\star$ is minimal. But what can we say about the individual features? We could say that the whole set bears responsibility but individual features do not, or that any feature in a cause set is partially and equally accountable. Both approaches have issues, as they imply the features are equally responsible. This conflicts with our intuition: OBCB did not evaluate Alice's credit score, so her denial was caused by her gender.

The structural pattern is what the actual-causality literature calls *preemption* (sometimes *undercutting*): two distinct causal routes might each have been sufficient to produce the outcome, but only one is actually engaged, with the other prevented from operating. Alice's gender preempted her credit by preventing the check from happening, so credit had no chance to act. The closely related notion of *over-determination* covers cases in which multiple sufficient causes coexist without one preempting the other (e.g., two simultaneous fatal shots). Both phenomena defeat naive but-for tests, since intervening on any one of the candidate causes leaves the outcome unchanged.[7]

Prior work suggests a way out of this difficulty: hold some of the context in which counterfactuals are evaluated fixed, calling such fixed variables *witnesses*. Let us focus on the mediator $\textit{check-failed}$ between $\textit{gender}$ and $\textit{loan}$. If we fix this variable at the factual value by intervening, then counterfactually, Alice would have been granted the loan if she were male. However, there is no context fixed at the actual value such that, had her credit score been higher, she would have been granted the loan. This motivates the following formalization:

Definition 5 (Actual cause)**.** *Let $\mathbf{S} \subseteq \mathbf{V}$ be a suspect pool and $\mathbf{W} \subseteq \mathbf{V}$ a witness pool, observed in $(M, \mathbf{u})$ at factual values $\mathbf{S} = \mathbf{s}^\star$ and $\mathbf{W} = \mathbf{w}^\star$. A cause set $\mathbf{C} \subseteq \mathbf{S}$ at its factual setting $\mathbf{C} = \mathbf{c}^\star$ is an* actual cause *of $Y = y^\star$ in $(M, \mathbf{u})$ if there is a witness set $\mathbf{T} \subseteq \mathbf{W}$ disjoint from $\mathbf{C}$ such that:*

- ***Factivity.*** *The cause, the witness pool, and the outcome are all observed at their factual values:*
$$(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^\star,\ \mathbf{W} = \mathbf{w}^\star,\ Y = y^\star.$$
- ***Witnessed necessity.*** *Holding $\mathbf{T}$ at its factual setting $\mathbf{t}^\star$ while intervening on the cause overturns the outcome, for some alternative $\mathbf{c}'$ distinct everywhere from $\mathbf{c}^\star$:*
$$(M, \mathbf{u})\big[\mathbf{C} \leftarrow \mathbf{c}',\ \mathbf{T} \leftarrow \mathbf{t}^\star\big] \not\models Y = y^\star.$$
- ***Minimality.*** *No proper subset of $\mathbf{C}$ satisfies the two conditions above.*

This definition extends but-for causality with a witness set $\mathbf{T}$ that is fixed to its factual value by an intervention. Under this definition, to determine the cause of an outcome, we need to find a cause set $\mathbf{C}$, alternative values $\mathbf{c}'$ (trivial if the variables are binary and less so if they are categorical or continuous), and a witness set $\mathbf{T}$. $\mathbf{C}$ and $\mathbf{T}$ are drawn from sets of size at most $2^{|\mathbf{V}|}$, so a brute-force computation is expensive.

Putting computational difficulties aside, the deeper limitation of the construction so far is that it lives entirely in deterministic models. Real-world attribution problems are typically probabilistic: policies apply with some probability, mechanisms occasionally fail or are overridden, and the analyst usually wants to reason about the unit-level outcome in the presence of this noise. Lifting actual causality into that setting is not straightforward, and is one of the reasons we turn next to a different family of attribution tools.

2.2 Probability of Necessity and Sufficiency

To bring probability into the picture, we now consider a stochastic variant of OBCB in which the bank's gendered policies apply only with some probability: women are checked only sometimes, men are skipped only sometimes, and even when the check happens its outcome is occasionally overridden by loan-officer discretion. This is the natural setting for Pearl's probabilities of necessity, sufficiency, and their conjunction (PN, PS, PNS), which we introduce in this subsection. As we will see, these scale gracefully into the stochastic regime but, without an analogue of the witness mechanism, mis-attribute Alice's denial.

[7] The canonical preemption example: Jim is both stabbed and poisoned, and dies. Stabbing works quickly, so it kills Jim before the poison takes effect, and the stabbing caused his death. But-for clauses fail to break the symmetry here as well: removing the stabbing leaves Jim dead by poison, and removing the poison leaves him dead by stab.

Example 6 (OBCB, stochastic)**.** *The bank performs credit checks for a randomly selected 20% of women and skips checks for a randomly selected 10% of men. Unchecked applicants are rejected. Furthermore, if a man's credit check fails, he is nevertheless accepted 5% of the time and if a woman's credit check succeeds, she is denied 10% of the time. We assume that the population is evenly divided into male and female and good and bad credit. Formally, the model, using 1 for male and 1 for good credit, is:*

$$
\textit{loan-prob}(\textit{gender}, \textit{check-failed}) = \begin{array}{c|cc} & \textit{check-failed}{=}0 & \textit{check-failed}{=}1 \\ \hline \textit{gender}{=}0 & 0.9 & 0 \\ \textit{gender}{=}1 & 1 & 0.05 \end{array}
$$

$$
\begin{aligned}
\textit{credit} &\sim \mathrm{Bern}(0.5) \\
\textit{gender} &\sim \mathrm{Bern}(0.5) \\
\textit{check} \mid \textit{gender} &\sim \mathrm{Bern}\big((1-\textit{gender})\cdot 0.2 + \textit{gender}\cdot 0.9\big) \\
\textit{check-failed} &= \textit{check}\cdot(1-\textit{credit}) \\
\textit{loan-if-checked} \mid \textit{gender}, \textit{check-failed} &\sim \mathrm{Bern}\big(\textit{loan-prob}(\textit{gender}, \textit{check-failed})\big) \\
\textit{loan} &= \textit{loan-if-checked}\cdot \textit{check}
\end{aligned}
$$

The witness mechanism works well for deterministic models, but actual causality in its standard form offers no straightforward way to handle probabilistic ones. Before extending the witness idea to the stochastic setting, it is instructive to examine a probabilistic approach that scales more naturally — Pearl's probabilities of necessity and sufficiency [Pearl, 2022] — and to ask what it would miss without witnesses. This will let us identify what probabilistic machinery is needed, and what context-sensitivity is lost when witnesses are absent, motivating the synthesis that PCI provides. The following definitions allow for probabilistic models, but assume that all variables are Boolean (PCI will relax this assumption, allowing for continuous variables). Pearl considers two ways that a cause may relate to an effect: it may be necessary or sufficient for the effect. A cause is necessary if, counterfactually absent the cause, the effect is unlikely to occur; it is sufficient if the effect is counterfactually likely in the presence of the cause. We now formalize these ideas and discuss them in the context of the running example.

Definition 7 (Probability of Necessity)**.** *Let the probability of necessity be the probability that the outcome $Y = y$ would not have occurred in the absence of $C = c^\star$, given that both occurred:*

$$PN(C = c^\star, Y = y^\star) = P(Y_{c'} \neq y^\star \mid C = c^\star, Y = y^\star)\,,$$

where $Y_{c'}$ is the value of Y after intervening on C with $c' \neq c^\star$.[8]

Strictly speaking, the probability of necessity is less specific to a particular situation than the actual cause test discussed above. For example, we can consider the probability that being male causes someone to fail to get a loan, but we can't consider the probability that being male causes Bob specifically (who we know has bad credit) to fail to get a loan. We might try to patch this up by conditioning, for the time being (while also reporting population-level values that follow the definitions more closely).

We illustrate the computation for Alice's gender. Alice is factually $\textit{gender}$ = female, $\textit{credit}$ = bad, $\textit{loan} = F$; the counterfactual intervenes to set $\textit{gender}$ = male while leaving $\textit{credit}$ = bad. Tracing the male-intervention branches through the model (Appendix A, §A.1) gives $PN(\textit{gender} =$ female, $\textit{loan} = F \mid$ Alice$) = 0.9 \cdot 0.05 = 0.045$. Computing the analogous quantities for the remaining (variable, person) pairs gives the individual-level values shown in Table 1; the table also lists population-level values for comparison.

The PN scores partially align with intuition: Bob's bad credit is high ($PN = 0.90$), and his gender score is zero, matching the reading that his gender played no role. But Alice's PN values do not separate gender (0.045) from credit (0.18) particularly well, and Alice's gender PN is only marginally

[8] Read in twin-network fashion: one draws an exogenous noise $\mathbf{u} \sim P_{\mathbf{U}}$, computes the factual Y under the unintervened structural equations and the counterfactual $Y_{c'}$ under the intervened equations, and conditions on the joint event $\{C = c^\star, Y = y^\star\}$. Both Y and $Y_{c'}$ live on the same probability space and share the same $\mathbf{u}$.

above Bob's, not the sharp distinction we would want, given that the bank never even evaluated Alice's credit. PN cannot register this asymmetry: it asks counterfactually what would happen under intervention without any context about which causal paths were active. For the full picture we need a second quantity, the probability of sufficiency.

Definition 8 (Probability of Sufficiency)**.** *Let the probability of sufficiency be the probability that the outcome $Y = y^\star$ would have occurred in the presence of $C = c^\star$, given that neither $C = c^\star$ nor $Y = y^\star$ occurred:*

$$PS(C = c^\star, Y = y^\star) = P(Y_{c^\star} = y^\star \mid C = c',\ Y \neq y^\star)$$

where $c' \neq c^\star$ and $Y_{c^\star}$ is the value of Y after intervening on C with $c^\star$.

To query whether the factual cause was sufficient, we condition on the counterfactual: we assume the cause did *not* take its factual value and the outcome did *not* occur, then intervene to restore the factual cause value and ask whether the outcome now occurs. Intuitively, this asks: if the cause had been absent and things had gone differently, would reinstating the cause have brought the outcome about? Let's look at the resulting values.

We illustrate PS for Alice's gender. Following the same convention as for PN, the individual-level computation additionally conditions on Alice's credit ($credit =$ bad): we fix her credit but flip the cause and outcome to their non-factual values ($gender =$ male, $loan = T$), then intervene to restore $gender =$ female and ask whether the loan is again denied. Both check branches yield $loan = F$ (Appendix A, §A.2), so $PS(gender = \text{female}, loan = F \mid \text{Alice}) = 1$. The remaining individual values, along with population-level values for comparison, appear in Table 1. Two cases warrant comment: Bob's gender PS is undefined because the required conditioning event ($gender =$ female, $credit =$ bad, $loan = T$) has probability zero in the model; Bob's credit PS is 0.955 rather than 1 because the model includes a 5% exception rate for males who fail the credit check.

To somehow combine insights provided by the two scores, we would like to find causes that are both necessary and sufficient. This notion is formalized as follows:

Definition 9 (Probability of necessity and sufficiency)**.** *Let the probability of necessity and sufficiency be the probability that the outcome $Y = y^\star$ would have occurred in the presence of $C = c^\star$ and that it would not have occurred under the intervention $C = c'$:*

$$PNS(C = c^\star, Y = y^\star) = P(Y_{c^\star} = y^\star, Y_{c'} \neq y^\star).$$

This quantity factorises into a weighted sum of PN and PS by the consistency rule of counterfactuals [Pearl, 2009, §9.2]:

$$PNS(C = c^\star,\ Y = y^\star) = P(C = c^\star,\ Y = y^\star)\ PN(C = c^\star,\ Y = y^\star) + P(C = c',\ Y \neq y^\star)\ PS(C = c^\star,\ Y = y^\star).$$

Table 1: PN, PS, and PNS for Alice and Bob in Example 6, with $loan = F$ as the outcome. "Population" values condition only on the cause–outcome pair prescribed by the definition; "Individual" values additionally condition on the subject's other factual variable (e.g. for Alice's gender, on $credit =$ bad). A dash marks an entry that is undefined because the conditioning premises are inconsistent with the model.

		Alice			Bob		
Variable	Level	PN	PS	PNS	PN	PS	PNS
$gender$	Population	0.43	0.83	0.39	0.02	0.10	0.01
$gender$	Individual	0.045	1.00	0.045	0	—	0
$credit$	Population	0.53	0.96	0.52	0.53	0.96	0.52
$credit$	Individual	0.18	1.00	0.18	0.90	0.955	0.86

The population- and individual-level PNS values appear in the rightmost column group of Table 1. These values capture the relative causal impact of $gender$ and $credit$ but are not entirely satisfying. Table 1 reveals the problem directly: for Alice, $PNS(gender) = 0.045 < PNS(credit) = 0.18$, so PNS ranks credit as more causally responsible than gender. Yet the bank never evaluated Alice's credit: the $check$ variable was F throughout. PNS has no mechanism to register this: it asks counterfactually what would happen if credit were different, without fixing $check\text{-}failed$, the mediating variable that blocked credit's causal path.

As an attempt to help Alice better understand her particular situation, we might consider a modification of PNS with additional conditioning:

$$PNS_c(C = c^\star, Y = y^\star \mid Z = z^\star) = P(Y_{c^\star} = y^\star, Y_{c'} \neq y^\star \mid Z = z^\star)$$

Using this definition, we can compute $PNS_c(\mathit{gender} = \text{female}, \mathit{loan} = F \mid \mathit{credit} = \text{bad}) = 0.045$ and $PNS_c(\mathit{credit} = \text{bad}, \mathit{loan} = F \mid \mathit{gender} = \text{female}) = 0.18$. Conditioning does not solve the problem described above (Alice's credit was in reality never evaluated) because the interventions override the mediating variable that tracks whether her credit has been checked. When we compute $PNS_c(\mathit{credit} = \text{bad}, \mathit{loan} = F \mid \mathit{gender} = \text{female})$, conditioning on gender being female is consistent with the factual, but when we then intervene to set credit to good, this intervention propagates downstream and changes $\mathit{check\text{-}failed}$, overriding what we learned from conditioning on Alice's factual outcome. The context we fixed is undone by the very intervention meant to probe it. We will not consider this use of conditioning further, and this observation motivates the need for intervention-based (as opposed to conditioning-based) context-sensitivity, which is what the witness mechanism provides.

The propagation problem is not specific to PNS. The two most familiar causal quantities in the ML and policy literature are the average treatment effect $\mathrm{ATE}(T) = \mathbb{E}[Y \mid do(T{=}1)] - \mathbb{E}[Y \mid do(T{=}0)]$ and the conditional version $\mathrm{CATE}(T \mid X{=}x) = \mathbb{E}[Y \mid do(T{=}1), X{=}x] - \mathbb{E}[Y \mid do(T{=}0), X{=}x]$. Computed on our stochastic OBCB, population ATE ranks credit (0.518) above gender (0.383), mirroring the population PN ranking. At the individual level for Alice, $\mathrm{CATE}(\mathit{credit} \mid \mathit{gender}{=}\text{female}) = 0.18$ and $\mathrm{CATE}(\mathit{gender} \mid \mathit{credit}{=}\text{bad}) = 0.045$. These are exactly Alice's individual PNS values: with binary outcomes and $PS = 1$ in the relevant branches, CATE for a single treatment collapses to PNS_c, so the same $\mathit{check\text{-}failed}$ propagation that undermined PNS_c undermines CATE. A second, sharper failure follows from CATE's inability to condition on the factual treatment value: $\mathrm{CATE}(\mathit{gender} \mid \mathit{credit}{=}\text{bad}) = 0.045$ is identical for Alice and Bob even though one's gender carried her denial outright while the other's merely opened the credit check that then rejected him. The structural limitation cuts across PNS, ATE, and CATE alike, which is why the witness mechanism in Section 3 is not just a fix for PNS but a general repair for the family of methods that average counterfactual effects without intervention-based context-sensitivity.

2.3 Path Forward

The two predecessors examined in this section have complementary strengths and blind spots. Halpern's actual causality is context-sensitive: by holding witnesses fixed at their factual values, it correctly diagnoses Alice's case, attributing her denial to gender rather than credit. But in its standard form it is restricted to deterministic models, it requires a brute-force search over witness sets, and it has no native treatment of continuous variables. Pearl's PN/PS/PNS make the converse trade-off. They scale naturally to probabilistic models, because every counterfactual reduces to an expectation under the noise distribution. But they have no analogue of the witness mechanism: every intervention propagates through the structural equations and overrides whatever mediator value the factual context fixed. Table 1 shows the symptom: for Alice, $PNS(\mathit{credit}) > PNS(\mathit{gender})$, even though the credit check was never performed.

Conditioning is not a workaround: we saw with PNS_c that context fixed by observation does not survive the subsequent intervention, since the credit intervention propagates downstream and overrides the $\mathit{check\text{-}failed}$ value that conditioning had fixed. Any replacement therefore needs intervention-based context-sensitivity, holding the relevant mediators fixed by intervention. A satisfactory replacement must combine three ingredients. First, from actual causality, the witness mechanism: a way to hold relevant context fixed by intervention so that the structural information about which causal paths were active survives the counterfactual evaluation. Second, from PN/PS/PNS, the probabilistic vocabulary: counterfactuals expressed as expectations under the noise distribution, so that the attribution lives on the same probability space as the model and naturally extends to continuous variables. Third, in place of brute-force enumeration over witness sets, an integral that can be estimated from samples even when the witness pool is large or the variables are continuous.

PCI provides this synthesis. To get some general intuition (we will revisit the example in more detail later), consider Alice's case once more, beginning with gender. PCI considers counterfactual worlds in which gender is flipped to male while a sampled witness set is held at its factual values by intervention. For Alice, the witness sets that drive the attribution are those containing $\mathit{check\text{-}failed}$:

when this variable is held fixed at F — the value recording that the check was never performed — flipping gender to male leads to loan approval. Unlike under PNS_c, the gender intervention does not override the mediator, because the witness intervention keeps it pinned. The counterfactual shift is large, and gender receives high attribution.

For credit, the same mechanism yields the opposite verdict. The witness sets that contain $check\text{-}failed{=}F$ now work against credit: holding that mediator fixed at F disconnects credit from the loan decision, because the bank's policy is to reject unchecked applicants regardless of credit. The counterfactual shift in those worlds is zero, and credit receives low attribution. The intuition that Alice's denial was a function of gender, not credit, is recovered.

Two features distinguish PCI from both predecessors. Unlike actual causality, it does not ask whether *some* witness set validates a counterfactual dependency; it integrates over a distribution of witness sets, weighting each by its probability. This replaces combinatorial enumeration with statistical estimation. Unlike PN/PS/PNS, it inherits the context-sensitivity of actual causality through this integration: causally inert variables are downweighted automatically, because almost every witness set leaves them disconnected from the outcome, while variables that drive the outcome are upweighted, because almost every witness set leaves them connected. The formal definitions in Section 3 make this expectation precise.

Before moving to the formal development, we flag the closest neighbour of PCI in the existing literature, so the later comparison does not come out of nowhere: Pearl's *probability of actual causation* [Pearl, 2009, Def. 10.3.5], the posterior probability — given the observed evidence — that the noise lands in a state for which the actual-cause verdict holds. Because PCI proposes a seemingly similar bridge from structural information to a probabilistic score, and the two give the same rankings in typical simple cases, we defer a precise side-by-side comparison to Section 8, once enough machinery is on the page. (The formula and its Bayes-rule reading are stated there, in Eq. (20).)

With the limitations of both predecessors in mind, we turn in Section 3 to the formal development. The next section introduces PCI piece by piece — PSCM scaffolding, the variable selection distribution Γ, the alternative-value distribution Δ, the joint necessity-sufficiency measure, and the user-chosen causal impact function ci — and verifies on the same Alice/Bob example that the resulting attribution satisfies all seven desiderata stated at the start of that section.

3 Causal Impact: Definitions

Explanation and causal attribution seek to connect a potentially causal *event* $X = x$ to an outcome event $Y = y$.[9] In this section, we present a formal development. To ground it, we return to Example 6 and use Alice and Bob as running illustrations throughout this section. Both are rejected for a loan; the question is which of their attributes ($gender$ and $credit$) was causally responsible for each rejection, and to what degree. While the example is discrete, the machinery applies to continuous and mixed-type variables; a continuous illustration is given in Sections 4.3 and 5.

Roadmap of this section. The development proceeds in five steps: (1) write down a list of seven *desiderata* that any responsibility attribution for Alice and Bob should satisfy, so that the formal construction can be judged against an explicit target; (2) introduce the probabilistic structural causal model (PSCM) and interventions, the substrate on which everything else rests; (3) introduce the two distributions PCI exposes to the user: the variable selection distribution Γ over suspect/witness pairs and the alternative-value distribution Δ over counterfactual cause settings; (4) combine them into the joint necessity-sufficiency measure $P_k^{s,n}$ and the user-chosen causal impact function ci whose expectation is the PCI score, and show how the choice of ci generalises the framework from binary to continuous outcomes; (5) compute the score on the OBCB example, verify that all seven desiderata hold, and contrast PCI's necessity/sufficiency decomposition with Pearl's PN/PS/PNS. The closing paragraph then points forward to Section 4, which positions PCI against SHAP and Causal SHAP.

Let $R(V \rightsquigarrow loan \mid \text{person})$ denote the causal responsibility of variable V for the loan outcome at the individual level (R is a placeholder for any attribution score and should not be identified

[9] This is quite different from type-level causal queries, where one is interested in some group-level attribution, as when one estimates average treatment effect or conditional average treatment effect.

with a specific formal quantity; PCI interprets R as $\mathbb{E}[ci(Y^s, Y^n, y^\star)]$, to be defined below, but the desiderata are method-agnostic). Attribution methods may produce signed values; the desiderata are stated in terms of absolute magnitudes $|R(\cdot)|$, allowing for directionality:

D-A1 $|R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Alice})| > 0$

D-A2 $|R(\mathit{credit} \rightsquigarrow \mathit{loan} \mid \text{Alice})| > 0$

D-A-rank $|R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Alice})| > |R(\mathit{credit} \rightsquigarrow \mathit{loan} \mid \text{Alice})|$

D-B1 $|R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Bob})| > 0$

D-B2 $|R(\mathit{credit} \rightsquigarrow \mathit{loan} \mid \text{Bob})| > 0$

D-B-rank $|R(\mathit{credit} \rightsquigarrow \mathit{loan} \mid \text{Bob})| > |R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Bob})|$

D-comp $|R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Alice})| > |R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Bob})|$

D-A1 and **D-A2** both require non-zero attribution for Alice because two distinct causal routes were available: gender filtered her out at the checking stage without her credit ever being evaluated, while bad credit would have mattered had she been checked. Neither is causally inert. **D-A-rank** is the substantive constraint: because Alice never underwent a credit evaluation, gender's causal role is immediate, whereas credit's is purely hypothetical. A method that ranks credit above gender for Alice is misattributing the cause of her rejection.

For Bob, **D-B2** reflects that the credit check occurred and directly produced the rejection: the causal chain $\mathit{credit} \to \mathit{check\text{-}failed} \to \mathit{loan}$ was activated. **D-B1** requires strictly positive attribution for gender: being male gave Bob a 90% probability of being checked, so gender opened the evaluation that rejected him. This indirect causal role is non-zero; a method returning exactly zero fails to capture it. **D-B-rank** corresponds to the intuition that credit is the proximate cause. **D-comp** captures the cross-individual asymmetry that is hardest for naive methods to express: gender is more causally responsible for Alice's rejection than for Bob's, because for Alice gender=female was sufficient on its own to produce the rejection — no evaluation took place, so credit was never a factor — while for Bob gender=male merely opened the evaluation that credit=bad then caused. We will verify at the end of this section that PCI satisfies all seven desiderata, while PN, PS, PNS fail some of them. Later, we will also revisit this example and its desiderata in the context of SHAP and Causal SHAP (Section 4).

We work within the context of probabilistic structural causal models [Pearl, 2009].

Definition 10 (Probabilistic Structural Causal Model (PSCM))**.** *A probabilistic structural causal model is a tuple*

$$M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle,$$

where $\mathbf{V} = \{V_1, \ldots, V_n\}$ *is a finite set of endogenous variables,* $\mathbf{U} = \{U_1, \ldots, U_n\}$ *is a set of exogenous variables, and* $\mathbf{F} = \{f_1, \ldots, f_n\}$ *is a set of structural assignments*

$$V_i := f_i(\mathbf{Pa}_i, U_i), \qquad i = 1, \ldots, n,$$

with $\mathbf{Pa}_i \subseteq \mathbf{V} \setminus \{V_i\}$ *denoting the parents of* V_i*. The distribution* $P_{\mathbf{U}}$ *is a probability measure on* $\operatorname{dom}(\mathbf{U})$*, and the assignments in* $\mathbf{F}$ *are assumed to be acyclic and to have a unique solution for* $\mathbf{V}$ *given* $\mathbf{U}$*.*[10]

A PSCM encodes which variables influence which others, through what mechanisms, and subject to what randomness. The structural equations specify each variable as a deterministic function of its causal parents and local noise; the exogenous distribution captures all irreducible uncertainty in the system. Together they make every counterfactual question well-posed and computable.

The OBCB model (Example 6) is a PSCM with endogenous variables

$$\mathbf{V} = \{\mathit{gender}, \mathit{credit}, \mathit{check}, \mathit{check\text{-}failed}, \mathit{loan}\},$$

[10] Working within the PSCM framework commits the practitioner to fully specified structural assignments $\mathbf{F}$ and an exogenous distribution $P_{\mathbf{U}}$, analogous to how Bayesian inference requires a likelihood and a prior. This is precisely what enables PCI to yield individual-level counterfactual attributions rather than population-level bounds, and once $\mathbf{F}$ and $P_{\mathbf{U}}$ are specified, sampling from $P_{\mathbf{U}}$ is a routine probabilistic inference operation.

and exogenous variables

$$\mathbf{U} = \{U_{gender}, U_{credit}, U_{check}, U_{loan}\}$$

which induce independent Bernoulli noise.

The structural assignments in the case at hand take the form $V_i = f_i(\mathbf{Pa}_i, U_i)$ promised by the PSCM definition above, with indicator-threshold mechanisms against uniform exogenous noise:

$$\begin{aligned}
gender &= \mathbb{1}[U_g \leq \alpha_g],\\
credit &= \mathbb{1}[U_c \leq \alpha_c],\\
check &= \mathbb{1}\big[U_{ck} \leq \gamma_F(1 - gender) + \gamma_M\, gender\big],\\
check\text{-}failed &= check(1 - credit),\\
loan\text{-}if\text{-}checked &= \mathbb{1}\big[U_\ell \leq \text{loan-prob}(gender, check\text{-}failed)\big],\\
loan &= loan\text{-}if\text{-}checked \cdot check,
\end{aligned}$$

with $U_g, U_c, U_{ck}, U_\ell \overset{\text{iid}}{\sim} \text{Unif}[0,1]$ as the exogenous noise. For the running example we take $\alpha_g = \alpha_c = 0.5$, $\gamma_F = 0.2$, $\gamma_M = 0.9$, and the values of loan-prob given in Example 6.

Definition 11 (Interventions)**.** *Given a model $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$, an intervention on variables $\mathbf{X} = \{X_1, \ldots, X_k\} \subseteq \mathbf{V}$ with values $\mathbf{x} = (x_1, \ldots, x_k)$ replaces the structural assignments for X_i by $X_i := x_i$, where $i = 1, \ldots, k$. We denote this intervention by either $[\mathbf{X} \leftarrow \mathbf{x}]$ or $do(\mathbf{X} = \mathbf{x})$.*

An intervention surgically replaces a variable's structural equation with a fixed constant, disconnecting it from its usual causes. The rest of the model propagates downstream as normal under the new assignment; the exogenous noise is unchanged.

In the OBCB model, $\text{do}(gender = 1)$ replaces Alice's gender assignment with male, leaving all other structural equations intact. The downstream effect propagates: $check$ now draws from $\text{Bern}(0.9)$ rather than $\text{Bern}(0.2)$, so Alice would be checked in 90% of noise realizations instead of 20%. This is the counterfactual world in which we ask whether her loan outcome would have differed.

Because PCI supports arbitrary variable types, it will be helpful to define a measure theoretic object corresponding to interventional outcomes. Because outcomes are a deterministic function of exogenous noise, this will be a Dirac measure.

Definition 12 (Pointwise Interventional Law)**.** *Let $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ be a probabilistic structural causal model and let $Y \in \mathbf{V}$. For an intervention $\text{do}(\mathbf{X} = \mathbf{x})$, let $Y' : \text{dom}(\mathbf{U}) \to \text{dom}(Y)$ denote the measurable map induced by the modified structural assignments. The* pointwise interventional law *of Y at fixed $\mathbf{u} \in \text{dom}(\mathbf{U})$ is the probability measure on $\text{dom}(Y)$ defined by*

$$P(Y \in A \mid \mathbf{u}, \text{do}(\mathbf{X} = \mathbf{x})) \;:=\; \mathbb{I}\{Y'(\mathbf{u}) \in A\} \;=\; \delta_{Y'(\mathbf{u})}(A), \qquad A \subseteq \text{dom}(Y).$$

The pointwise interventional law records the deterministic outcome of Y under $\text{do}(\mathbf{X} = \mathbf{x})$ for a fixed noise realization $\mathbf{u}$: since the modified structural equations make Y a function of $\mathbf{u}$ alone, its distribution collapses to a point mass. Integrating over $P_{\mathbf{U}}$ then recovers the full interventional distribution.

We define a *causal set* to be a set of assignments to the variables of a probabilistic structural causal model.

Definition 13 (Causal set)**.** *Given a model $\langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$, a causal set is a set of pairs $X = x$ where $X \in \mathbf{V}$ and $x \in dom(X)$.*

Informally, a causal set is a list of specific variable-value assignments, a precise statement of which variables are set to which values. It is the basic unit of intervention: from a causal set one can read off exactly what is being manipulated or held fixed.

In PCI, causal sets play four distinct roles, two on the cause side and two on the context side:

Suspects $\mathbf{S}$. The full set of variables suspected to bear on the outcome. Subsets of $\mathbf{S}$ will be sampled and intervened on.

Active suspects $\mathbf{C} \subseteq \mathbf{S}$. The specific subset intervened on in a given search step, either to alternative values (necessity regime) or to factual values (sufficiency regime).

Witnesses $\mathbf{W}$. The full set of context variables that may be held fixed at their factual values. Subsets of $\mathbf{W}$ will be sampled and held fixed.

Active witnesses $\mathbf{T} \subseteq \mathbf{W}$. The specific subset held fixed in a given search step. To avoid contradicting the active intervention, $\mathbf{T}$ is pruned so that $\mathbf{T} \cap \mathbf{C} = \emptyset$.

Why prune $\mathbf{T}$ and not $\mathbf{C}$? The conflict on $\mathbf{T} \cap \mathbf{C}$ has to be resolved in one direction or the other, and pruning the witnesses is the right choice for three reasons. First, $\mathbf{C}$ is the object of the test: it encodes the causal claim being evaluated, while $\mathbf{T}$ is the auxiliary device that holds context fixed. Pruning $\mathbf{C}$ would silently change the claim under test; pruning $\mathbf{T}$ leaves the claim intact and merely weakens the context held fixed — a weaker probe of the same hypothesis, not a different one. Second, the convention extends cleanly to the limit $\mathbf{C} = \mathbf{V}$, where every variable is a suspect: the pruned $\mathbf{T}$ collapses to $\emptyset$ and we recover the unconditional but-for test, the right baseline behaviour. Pruning $\mathbf{C}$ instead would leave this limit undefined whenever any witness is sampled. Third, the convention inherits directly from Halpern's actual-cause definition (Definition 5), which makes $\mathbf{T} \cap \mathbf{C} = \emptyset$ part of the definition; pruning post-sampling is the operational implementation of that constraint and lets us specify Γ on the unconstrained product $2^{\mathbf{S}} \times 2^{\mathbf{W}}$ rather than baking disjointness into the prior.

To evaluate the causal impact of a variable on an outcome, we consider two interventional regimes: the *necessity regime*, in which we intervene on the variables $\mathbf{C} \subseteq \mathbf{S}$ to have an alternative value, and the sufficiency regime, in which we intervene so that $\mathbf{C}$ takes on factual values. Moreover, we hold some elements $\mathbf{T} \subseteq \mathbf{W}$ of the context fixed (i.e., intervened on to retain factual values, despite intervention on $\mathbf{C}$). This is exactly what makes the necessity component *context-sensitive* in Halpern's sense [Halpern, 2016]: the alternative cause is evaluated against a held-fixed witness configuration, not in isolation. We will reuse this terminology throughout: "context-sensitive necessity" refers to the necessity regime evaluated under an active witness set $\mathbf{T}$. When looking for an explanation, we perform a search across possible cause sets and possible contexts. Complete model samples in these regimes will be called *necessity worlds* and *sufficiency worlds*, respectively.

The intuition is that the suspect set $\mathbf{S}$ names every variable that could plausibly have caused the outcome; the witness set $\mathbf{W}$ names context variables to be selectively held fixed, isolating the causal channels under investigation. Intervening on suspects tests what would have happened under an alternative cause in the necessity worlds and under the factual values in the sufficiency worlds; fixing witnesses blocks confounding paths that would otherwise absorb the effect. Responsibility attribution increases when the outcome is far from the factual value in the necessity world and decreases when it is far from the factual value in the sufficiency world.

For both Alice (female, bad credit, loan denied) and Bob (male, bad credit, loan denied) we set $\mathbf{S} = \{\mathit{gender}, \mathit{credit}\}$ and $\mathbf{W} = \{\mathit{check\text{-}failed}\}$. The witness $\mathit{check\text{-}failed}$ records whether the applicant was checked and had bad credit; holding it at its factual value isolates the causal path from gender or credit to loan independently of the credit-check bottleneck. Without this witness, intervening on credit for Alice would leave 80% of noise realizations unaffected (those in which she is never checked), making credit's causal role invisible. The witness makes the bottleneck explicit and manipulable.

Moving on with the general framework, we will be searching for causes by intervening on subsets of suspects and holding fixed subsets of witnesses. To make this search tractable, we need a way to prioritize which subsets to consider.

Definition 14 (Variable Selection Distribution)**.** *Let $\mathbf{X}$ be a finite set of random variables. A* variable selection distribution *Γ is a probability distribution on $2^{\mathbf{X}}$. We write $\mathbf{Y} \sim \Gamma$ to mean $\mathbf{Y}$ is a random subset of $\mathbf{X}$ drawn from Γ, with probability mass function $p^{\Gamma}(\mathbf{y}) = \mathbb{P}(\mathbf{Y} = \mathbf{y})$ for $\mathbf{y} \subseteq \mathbf{X}$.*

The variable selection distribution Γ encodes prior beliefs about which subsets of suspects are worth examining. A uniform distribution treats all suspect subsets as equally plausible; a cardinality-constrained choice focuses the search on interactions of a particular order, analogous to choosing which interaction terms to include in a regression. More complex distributions could encode domain knowledge about which variables are more likely to interact, or could be learned from data.

In particular, we will be using variable selection distributions for the powerset of causal suspects $\mathbf{S}$ and for the powerset of witnesses $\mathbf{W}$. One simple pattern of suspect and witness selection can be derived by setting size limits on the number of activated suspects or witnesses, and sampling uniformly otherwise.

Example 15 (Cardinality-Constrained Uniform Selection)**.** *Let* $\mathbf{X}$ *be a candidate set of random variables and let* $k \sim \text{Unif}\{J, \ldots K\}$ *with* $1 \leq J \leq K \leq |\mathbf{X}|$*. A variable selection* $\mathbf{Y} \sim \Gamma(2^{\mathbf{X}})$ *is sampled from a cardinality-constrained uniform selection distribution* $\Gamma(2^{\mathbf{X}})$ *with probability mass function*

$$p^{\Gamma}(\mathbf{Y}) \propto \mathbb{I}[|\mathbf{Y}| = k].$$

The parameters J *and* K *determine the cardinality range of the search; specific choices connect PCI to existing methods and enable efficient Monte Carlo approximation.*[11]

In PCI the variable selection distribution Γ is a probability mass function on $2^{\mathbf{S}} \times 2^{\mathbf{W}}$, governing the joint choice of an active suspect set $\mathbf{C} \subseteq \mathbf{S}$ and an active witness set $\mathbf{T} \subseteq \mathbf{W}$. The recommended construction samples suspects and witnesses separately and then enforces the constraint that no variable plays both roles in the same draw: pick a marginal Γ_s on $2^{\mathbf{S}}$ and a marginal Γ_w on $2^{\mathbf{W}}$, draw $\mathbf{C} \sim \Gamma_s$ and $\mathbf{T} \sim \Gamma_w$ independently, and *reject* any draw with $\mathbf{T} \cap \mathbf{C} \neq \emptyset$, resampling until the constraint holds. Equivalently,

$$p^{\Gamma}(\mathbf{C}, \mathbf{T}) \;\propto\; p_s^{\Gamma}(\mathbf{C})\, p_w^{\Gamma}(\mathbf{T})\, \mathbb{I}[\mathbf{T} \cap \mathbf{C} = \emptyset].$$

Throughout, Γ_s and Γ_w are abbreviations for the sampling marginals of this construction; the formal primitive in Definition 18 is the joint Γ.

For the OBCB analysis, with $|\mathbf{S}| = 2$, we use a Γ built from Γ_s uniform over the three non-empty subsets of $\mathbf{S}$ ($\{\textit{gender}\}$, $\{\textit{credit}\}$, $\{\textit{gender}, \textit{credit}\}$, each with probability $1/3$) and Γ_w uniform over $\{\emptyset, \{\textit{check-failed}\}\}$, each with probability $1/2$. Since no element appears in both $\mathbf{S}$ and $\mathbf{W}$, the rejection step is vacuous and Γ is exactly the product of marginals $p_s^{\Gamma} \cdot p_w^{\Gamma}$. In larger suspect sets the choice of cardinality range becomes substantive: sampling uniformly from the full powerset yields an expected intervention size of $|\mathbf{S}|/2$, which spreads attribution across large coalitions and may make individual feature contributions harder to isolate; we return to this issue in Section 9, where the cardinality cap is treated as a tunable hyperparameter and a dynamic-set-sizes ablation measures the empirical cost of bounding it.

The choice of Γ is a design decision that shapes the search process and the resulting attributions. As a default, a uniform Γ over the full powerset is the least-informative starting point, assigning equal weight to all subset sizes, analogous to including all interaction orders in a regression. There is no universally principled criterion for restricting the cardinality range; the honest advice is to treat J and K as explicit design choices. Two practical considerations bear on this choice, however: with $K = 1$, attribution scores are based on singleton interventions and straightforward to communicate to stakeholders; higher K averages over joint interventions on X_k with other suspects simultaneously, which is harder to explain and, in large or structurally complex models, may amplify estimation noise rather than capturing genuine higher-order causal structure. For applications where attribution has serious consequences, we recommend a cardinality sensitivity analysis: if rankings are stable as K increases from 1 to $|\mathbf{S}|$, the simpler choice is supported; if they diverge, higher-order interactions

[11] When $J = 1$ and $K = |\mathbf{X}|$, this distribution coincides with the Shapley weighting kernel, which samples coalition size uniformly and then selects a coalition uniformly at that size [Shapley, 1953]. The parameters J and K allow the practitioner to restrict the search to cause sets of a specific cardinality range: setting $J = K = 1$ confines attribution to individual variables, while $J = 2, K = 3$ focuses on small multi-variable interactions. Moreover, whereas exact Shapley computation requires summing over all $2^{|\mathbf{X}|}$ coalitions, PCI treats Γ as a generative model: approximation proceeds by sampling subsets from Γ and outcomes from $P_{\mathbf{U}}$, making Monte Carlo estimation a native feature of the framework rather than an external algorithmic layer. We note that the weighting alone does not distinguish PCI from SHAP at $J = 1$, $K = |\mathbf{X}|$: the subset weights are identical. The distinction lies in the integrand, and it matters for what is being evaluated and what results follow. SHAP averages marginal contributions using the observational conditional $\mathbb{E}[f(\mathbf{X}_{\mathcal{S}} = \mathbf{x}_{\mathcal{S}}, \mathbf{X}_{\bar{\mathcal{S}}})]$, marginalising over the empirical distribution of non-suspect features; this means a variable that is merely correlated with a cause, but not itself causally active, can receive positive attribution. PCI instead evaluates counterfactual outcomes $P(Y \in A \mid \mathbf{u}, \text{do}(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{t}^{\star}))$ computed via the structural model, so attribution flows only along genuine causal paths. The witness mechanism adds a further distinction: by holding intermediate variables fixed, PCI can resolve overdetermination and undercutting scenarios where SHAP assigns equal or incorrect attribution. The choice of J and K remains a design decision; practitioners can set $J = K = 1$ for single-variable attribution or allow higher-order interactions as discussed above. A detailed comparison of PCI, SHAP, and Causal SHAP on concrete examples is given in Section 4.

are materially relevant and the full powerset should be used. The key advantage of PCI is that this commitment is made transparently rather than absorbed into an implicit default.[12]

Causal attribution hinges on questions such as "if X had, contrary to fact, been fixed to $x' \neq x$ instead of x, what would the outcome have been?" For non-binary causes, this is ambiguous. To resolve this in a way that does not require committing to a single, contrastive causal event [Kawakami et al., 2024], we rely on a *distribution* over alternative values.

Definition 16 (Alternative Value Distribution)**.** *Let* $\mathbf{X}$ *be a set of random variables with domain* $\mathrm{dom}\,(\mathbf{X})$ *and let* $\mathbf{X} = \mathbf{x} \in \mathrm{dom}\,(\mathbf{X})$ *be a factual event. An* alternative value distribution $\Delta(\mathbf{x})$ *is a probability measure on* $\mathrm{dom}\,(\mathbf{X})$ *such that* $\mathbf{x} \notin \mathrm{supp}(\Delta(\mathbf{x}))$.

Informally, $\Delta(\mathbf{x})$ specifies what counts as a genuinely different value for a suspect variable. The support condition ensures no mass falls on the factual value $\mathbf{x}$ itself, so every sampled alternative is a real counterfactual departure rather than a noisy copy of what actually happened.

Three properties guide the choice of Δ. *Plausibility under* M*:* the alternative should be a value the structural model could plausibly produce given the context, so the necessity test interrogates a counterfactual the model has anything to say about. *Outcome-independence:* the distribution over $\mathbf{x}'$ should not be informed by the outcome whose causes we are explaining, on pain of circularity. *Context-sensitivity:* $\mathbf{x}'$ should reflect the rest of the factual scenario, so the necessity test is individual-level rather than population-level. The construction we use throughout is the structural-model posterior of $\mathbf{X}$ conditioned on upstream factual values, optionally excised near the factual value (Example 17 below); this is the natural choice satisfying all three.

Upstream-only conditioning operationalises the intuition shared with the descriptive-normality literature [Halpern and Hitchcock, 2015, Icard et al., 2017]: factual upstream values play the role of the *default* against which counterfactual alternatives are graded. PCI does not aim to resolve the specific counterexamples motivating that literature; the link is at the level of *why* a non-uniform Δ matters at all. Since Δ is user-pluggable, other constructions are available when domain knowledge motivates them: a uniform distribution over $\mathrm{dom}\,(\mathbf{X})$ trades plausibility for breadth; the empirical marginal $p(\mathbf{x}')$ trades context-sensitivity for simplicity; a fully conditional posterior on every observable (including downstream evidence) trades outcome-independence for sharper conditioning. Each reads as an informed relaxation of one of the three properties above, not an unconsidered default — exactly the user-facing knob the framework is designed to expose.

One natural alternative value distribution results from taking the one currently embodied in the model and excising it around the factual setting. Then, the semantics for "alternative value" means "a sufficiently different value, following the original distribution otherwise". In the discrete case, of course, the analogue is simply the probability mass function conditioned on the factual value not holding.

Example 17 (ϵ-Excised Distribution)**.** *Assume* $\mathbf{X}$ *is* $\mathbb{R}^d$*-valued with density* p*. Given a factual event* $\mathbf{X} = \mathbf{x}$*, define the* ϵ*-excised alternative value distribution* $\Delta_\epsilon(\mathbf{x})$ *as the probability measure on* $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ *with density*

$$p_{\Delta_\epsilon}(\mathbf{x}' \mid \mathbf{x}) \propto \begin{cases} 0, & \mathbf{x}' \in B_\epsilon(\mathbf{x}), \\ p(\mathbf{x}'), & \textit{otherwise}, \end{cases}$$

where $B_\epsilon(\mathbf{x})$ *denotes the* ϵ*-ball centered at* $\mathbf{x}$*. The support is therefore* $\mathrm{supp}(\Delta_\epsilon(\mathbf{x})) = \mathrm{supp}(p) \setminus B_\epsilon(\mathbf{x})$ *— not just the factual point* $\mathbf{x}$ *but its entire* ϵ*-neighbourhood is excluded.*

Informally, the alternative value distribution Δ operationalises what "counterfactually different" means for each suspect variable: it specifies which alternative values are considered when asking whether the outcome would have changed under an alternative cause.

[12] SHAP faces an analogous decision without flagging it as one: by averaging over all $2^{|N|}$ feature coalitions, plain SHAP and Causal SHAP commit to the maximum interaction order $K = |N|$ as an unstated default, with no diagnostic offered for whether that level of joint conditioning is right for a given model. Restricting SHAP to $K < |N|$ would break the efficiency axiom, so the choice is hardcoded into the framework rather than treated as a tunable. PCI surfaces the same decision as an explicit knob.

In the OBCB model both $gender$ and $credit$ are binary, so Δ is trivial. For Alice, the only alternative to $gender = 0$ (female) is $gender = 1$ (male), and the only alternative to $credit = 0$ (bad) is $credit = 1$ (good); for Bob, the only alternative to $gender = 1$ is $gender = 0$. Accordingly, $\Delta(gender = 0)$ is a point mass on $gender = 1$ and $\Delta(credit = 0)$ a point mass on $credit = 1$: no distributional choice is required. The binary case is exactly where Δ is unambiguous; it is for continuous suspects such as income or loan amount that the ϵ-excised distribution becomes essential, forcing an explicit commitment to what counts as a meaningfully different value.

Why ϵ-excision? Any method that asks "what would happen if $\mathbf{X}$ had been different" must operationalise what *different* means. Three natural choices for Δ are available. The observational marginal $p(\mathbf{x}')$ — as in kernel SHAP [Lundberg and Lee, 2017] — places mass near the factual value $\mathbf{x}$, conflating the necessity question ("what if $\mathbf{x}$ had genuinely differed?") with evaluating essentially the same setting. The fully conditional posterior $p(x'_k \mid \mathbf{x}_{-k})$ conditions on all other observed features, potentially including descendants of X_k, which imposes structural constraints that may rule out otherwise plausible counterfactuals, and still concentrates mass near x_k when features are correlated. The ϵ-excised distribution occupies the middle ground: it draws from the marginal informed by upstream variables only, excising the factual neighbourhood to guarantee genuine departure without downstream conditioning.

The choice parallels the Region of Practical Equivalence (ROPE) in Bayesian hypothesis testing [Kruschke, 2018]: just as ROPE forces an explicit decision about what difference is large enough to matter, ϵ-excision forces an explicit decision about what counterfactual is sufficiently far from the factual to count as genuinely alternative. The practitioner makes this commitment regardless of method; the only question is whether it is made transparently or absorbed into a default. When X is standardised to $\mathcal{N}(0, 1)$ the choice acquires a direct probabilistic interpretation: the excised mass $\Phi(x^\star+\epsilon)-\Phi(x^\star-\epsilon)$ is the fraction of the distribution ruled out as insufficiently different from $x^\star$. At $x^\star = 0$, $\epsilon = 0.2,\ 0.5,\ 0.8$ excise approximately $16\%, 38\%, 58\%$ of the distribution, corresponding to small, medium, and large effect thresholds in the sense of Cohen [1988]. Alternatively, ϵ can reflect measurement precision or a decision-relevant threshold; sensitivity analysis across ϵ values is advisable when the appropriate scale is unclear. For concrete examples of how SHAP's implicit choice leads to problematic attributions, see Section 4.

Where we are. The PSCM and intervention apparatus give us a notion of counterfactual outcome for any fixed noise realisation $\mathbf{u}$; Γ tells us which suspect/witness configurations to ask about; Δ tells us which non-factual values to substitute when evaluating necessity. What remains is to plug those distributions into the counterfactual mechanics and read out a number. The next two definitions do exactly that: first the joint necessity-sufficiency measure $P_k^{s,n}$, then its expectation against a user-chosen impact function ci, with the OBCB scores and the desiderata verification immediately after.

We are now ready to state the main definitions of PCI. The core idea is to measure two complementary things for a variable of interest X_k: (a) how much X_k's factual value contributes to producing $y^\star$, the *sufficiency* question, and (b) how much replacing X_k's value with alternatives would shift the outcome away from $y^\star$, the *necessity* question. We capture both by constructing two counterfactual outcome distributions conditioned on the same noise realization $\mathbf{u}$, then marginalising their product over $P_{\mathbf{U}}$. Because PCI is not constrained to particular variable types, we work in general measure-theoretic terms supporting combinations of continuous and discrete variables.

Definition 18 (Joint Necessity and Sufficiency Measure). *Let $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ be a probabilistic structural causal model and Y an outcome variable with domain $\operatorname{dom}(Y)$. Let p^Γ be the mass function of a variable selection distribution on $2^{\mathbf{S}} \times 2^{\mathbf{W}}$, and let $\Delta(\mathbf{s}^\star)$ be an alternative value distribution on $\operatorname{dom}(\mathbf{S})$. Let $(\mathbf{S}, \mathbf{W}) = (\mathbf{s}^\star, \mathbf{w}^\star)$ be a factual event.*

For a variable of interest X_k, let $2_k^{\mathbf{S}} := \{\mathbf{C} \subseteq \mathbf{S} : X_k \in \mathbf{C}\}$, and for each $\mathbf{C} \subseteq \mathbf{S}$ let $\Delta_{\mathbf{C}}(\mathbf{s}^\star)$ denote the pushforward of $\Delta(\mathbf{s}^\star)$ under the restriction map $\mathbf{s}' \mapsto \mathbf{s}' |_{\mathbf{C}}$.[13] *For any measurable $A \subseteq \operatorname{dom}(Y)$ and exogenous realization $\mathbf{u}$, define three sub-probability measures on $\operatorname{dom}(Y)$ (the first two) and $\operatorname{dom}(Y) \times \operatorname{dom}(Y)$ (the third).*

[13] For an assignment $\mathbf{x}$ to a collection of variables $\mathbf{X}$ and any subset $\mathbf{Y} \subseteq \mathbf{X}$, the restriction $\mathbf{x} |_{\mathbf{Y}}$ denotes the sub-assignment of $\mathbf{x}$ to the variables in $\mathbf{Y}$. Note that the restricted marginal $\Delta_{\mathbf{C}}(\mathbf{s}^\star)$ may place positive mass on $\mathbf{s}^\star |_{\mathbf{C}}$ even though $\Delta(\mathbf{s}^\star)$ excludes the joint factual $\mathbf{s}^\star$; the genuine-departure guarantee is on the joint alternative, not on each restricted coordinate.

(i) Sufficiency measure. *$P_k^s(\cdot \mid \mathbf{u})$ intervenes on each active suspect set $\mathbf{C}$ to its factual values $\mathbf{s}^\star|_{\mathbf{C}}$, holds the active witnesses $\mathbf{T}$ fixed, and takes the Γ-weighted sum of $P(Y \in A \mid \mathbf{u}, \mathrm{do}(\cdot))$ under factual cause restoration:*

$$P_k^s(A \mid \mathbf{u}) = \sum_{\substack{(\mathbf{C},\mathbf{T}) \\ X_k \in \mathbf{C}}} P(Y \in A \mid \mathbf{u}, \mathrm{do}(\mathbf{C} = \mathbf{s}^\star|_{\mathbf{C}}, \mathbf{T} = \mathbf{w}^\star|_{\mathbf{T}}))\; p^\Gamma(\mathbf{C}, \mathbf{T}). \tag{1}$$

(ii) Necessity measure. *$P_k^n(\cdot \mid \mathbf{u})$ instead integrates over alternative values $\mathbf{c}'$ for $\mathbf{C}$ (with $\mathbf{T}$ still pinned at factual values), and takes the Γ-weighted average over alternative cause values:*

$$P_k^n(A \mid \mathbf{u}) = \sum_{\substack{(\mathbf{C},\mathbf{T}) \\ X_k \in \mathbf{C}}} \int P(Y \in A \mid \mathbf{u}, \mathrm{do}(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{w}^\star|_{\mathbf{T}}))\, \Delta_{\mathbf{C}}(\mathbf{s}^\star)(d\mathbf{c}')\, p^\Gamma(\mathbf{C}, \mathbf{T}). \tag{2}$$

(iii) Joint necessity-sufficiency measure. *On product rectangles $A \times B \subseteq \mathrm{dom}\,(Y) \times \mathrm{dom}\,(Y)$, set*

$$P_k^{s,n}(A \times B) \;:=\; \int P_k^s(A \mid \mathbf{u})\, P_k^n(B \mid \mathbf{u})\, P_{\mathbf{U}}(d\mathbf{u}). \tag{3}$$

This rectangle-by-rectangle specification extends uniquely (by Carathéodory's extension theorem) to a sub-probability measure on the product σ-algebra of $\mathrm{dom}\,(Y) \times \mathrm{dom}\,(Y)$, with total mass $\Pr_\Gamma[X_k \in \mathbf{C}]^2 \leq 1$.

Informally: $P_k^s(\cdot \mid \mathbf{u})$ is the distribution of Y in the *sufficiency world*: restore the factual value of a candidate cause set $\mathbf{C}$ containing X_k, hold the witness set $\mathbf{T}$ at its factual values, and observe where the outcome lands. $P_k^n(\cdot \mid \mathbf{u})$ is the distribution of Y in the *necessity world*: replace $\mathbf{C}$'s values with alternatives drawn from Δ, hold $\mathbf{T}$ fixed, and ask how the outcome shifts. The sum over suspect-containing pairs $\{(\mathbf{C}, \mathbf{T}) : X_k \in \mathbf{C}\}$ weights each configuration by its Γ-mass: more focused interventions receive whatever weight the practitioner assigns them; configurations with $X_k \notin \mathbf{C}$ would contribute 0 to the attribution and are absent from the sum, so P_k^s and P_k^n are sub-probability measures with total mass $\Pr_\Gamma[X_k \in \mathbf{C}] \leq 1$ rather than renormalized expectations. In the OBCB example, for $X_k = \mathit{gender}$ under a noise realization in which Alice is credit-checked: the sufficiency world restores $\mathit{gender} =$ female while holding check-failed fixed; the necessity world substitutes $\mathit{gender} =$ male under the same noise.

The product decomposition in $P_k^{s,n}$ is legitimate because, conditional on $\mathbf{u}$, the sufficiency and necessity interventions operate in separate counterfactual regimes with no causal connection: $\mathrm{do}(\mathbf{C} = \mathbf{c}^\star)$ and $\mathrm{do}(\mathbf{C} = \mathbf{c}')$ act on the same noise but in distinct hypothetical worlds, so Y^s and Y^n are conditionally independent given $\mathbf{u}$, and $P(Y^s \in A,\, Y^n \in B \mid \mathbf{u}) = P_k^s(A \mid \mathbf{u})\, P_k^n(B \mid \mathbf{u})$.

With the necessity-sufficiency measure fully defined, we are ready to characterize the causal impact of an event X_k. In particular, we will proceed by defining an arbitrary *causal impact function*, whose expectation over the necessity-sufficiency measure determines the causal impact. We first proceed with a formal definition of the causal impact measure, followed by several intuitive candidates for the causal impact function.

Definition 19 (Causal Impact Function and Its Expectation)**.** *Let $y^\star \in \mathrm{dom}\,(Y)$ denote the factual outcome value, and let $Y^s, Y^n \sim P_k^{s,n}$. For any measurable function $ci : \mathrm{dom}\,(Y)^3 \to \mathbb{R}$, the* (expected) causal impact *of X_k is defined as*

$$\mathbb{E}[\, ci(Y^s, Y^n, y^\star)] \;:=\; \iint ci(y^s, y^n, y^\star)\, P_k^{s,n}(dy^s, dy^n). \tag{4}$$

The integral is taken against the sub-probability $P_k^{s,n}$ directly, not against its renormalisation; the symbol $\mathbb{E}[\cdot]$ is shorthand for this sub-probability integral throughout. Suspect configurations with $X_k \notin \mathbf{C}$ contribute zero mass to P_k^s and P_k^n (and hence to $P_k^{s,n}$), correctly capturing that those configurations make no claim on X_k's attribution.

Informally, ci is a user-defined scoring rule that translates the joint sufficiency–necessity measure into a single scalar. The choice is not arbitrary: the Absolute Difference example below is an instance of L_1 distance between outcomes, the same metric underlying Average Treatment Effect, Conditional Treatment Effect, and SHAP. The binary indicator score recovers this as a special case when outcomes are binary: indicators are L_1 on $\{0,1\}$. Practitioners with reasons to prefer L_2 or another distance measure may substitute it freely; the framework places no restriction on ci beyond measurability.

For the OBCB analysis we instantiate ci as the PNS binary score defined in the following example, with $y^\star = loan = 0$. This is the choice that produces the values in Table 2.

Example 20 (PNS for Binary Outcomes)**.** *Assume the outcome Y is binary with $\mathrm{dom}\,(Y) = \{0,1\}$, and let $y^\star = 1$ denote the factual outcome. Let Y^s and Y^n denote the sufficiency and necessity outcome variables. Define the causal impact score*

$$ci(y^s, y^n, y^\star) = \mathbb{I}\{y^n \neq y^\star\}\, \mathbb{I}\{y^s = y^\star\}.$$

Then the expected causal impact reduces to

$$\mathbb{E}[ci(Y^s, Y^n, 1)] = \mathbb{P}(Y^n = 0,\, Y^s = 1),$$

which conceptually coincides with Pearl's probability of necessity and sufficiency (PNS). The alignment is exact *when $\mathbf{S} = \{X_k\}$, $\mathbf{W} = \emptyset$, and $\mathrm{dom}\,(X_k) = \{0,1\}$ where, factually, $x_k^\star = 1$.*

We now illustrate the full computation for Alice's gender attribution using this ci. With $\mathbf{S} = \{gender, credit\}$, $\mathbf{W} = \{check\text{-}failed\}$, Γ_s and Γ_w uniform, Δ a point mass on the alternative binary value, and $y^\star = loan = 0$, the exogenous noise U_{check} falls into three regions:

- u_1 (prob. 0.2): Alice is checked regardless of gender; $check\text{-}failed = 1$ factually.
- u_2 (prob. 0.7): Alice is not checked as female but would be as male; $check\text{-}failed = 0$ factually.
- u_3 (prob. 0.1): Alice is not checked regardless of gender.

In all regions, restoring Alice's factual values always yields a denial, so $P_k^s(\{0\} \mid \mathbf{u}) = \frac{2}{3}$ everywhere (four suspect-containing (C,T) combinations each with $p^\Gamma(C,T) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$ — since $\mathbf{S} \cap \mathbf{W} = \emptyset$ here, rejection is vacuous and Γ is just the product of marginals — each contributing probability 1). The necessity measure varies.

In u_1, the witness holds $check\text{-}failed = 1$. The four (C,T) contributions, each weighted $\frac{1}{6}$, are:

- $\mathbf{C} = \{gender\}, T = \emptyset$: male, $check\text{-}failed = 1$ propagates structurally $\Rightarrow P(loan = 1) = 0.05$.
- $\mathbf{C} = \{gender\}, T = \{check\text{-}failed\}$: male, witness holds failure $\Rightarrow P(loan = 1) = 0.05$.
- $\mathbf{C} = \{gender, credit\}, T = \emptyset$: male and good credit, check passes $\Rightarrow P(loan = 1) = 1.0$.
- $\mathbf{C} = \{gender, credit\}, T = \{check\text{-}failed\}$: male and good credit, but witness holds failure $\Rightarrow P(loan = 1) = 0.05$.

$$P_k^n(\{1\} \mid u_1) = \tfrac{1}{6}(0.05 + 0.05 + 1.0 + 0.05) \approx 0.192.$$

In u_2, the witness holds $check\text{-}failed = 0$. The four contributions:

- $\mathbf{C} = \{gender\}, T = \emptyset$: male would be checked; $check\text{-}failed = 1$ propagates structurally $\Rightarrow P(loan = 1) = 0.05$.
- $\mathbf{C} = \{gender\}, T = \{check\text{-}failed\}$: male, witness holds $check\text{-}failed = 0$, check passes $\Rightarrow P(loan = 1) = 1.0$.
- $\mathbf{C} = \{gender, credit\}, T = \emptyset$: male and good credit, check passes $\Rightarrow P(loan = 1) = 1.0$.
- $\mathbf{C} = \{gender, credit\}, T = \{check\text{-}failed\}$: male and good credit, witness holds $0 \Rightarrow P(loan = 1) = 1.0$.

$$P_k^n(\{1\} \mid u_2) = \tfrac{1}{6}(0.05 + 1.0 + 1.0 + 1.0) \approx 0.508.$$

In u_3, male Alice is also not checked ($check = 0$ regardless of gender), so all necessity worlds yield denial: $P_k^n(\{1\} \mid u_3) = 0$.

Combining, with $ci(y^s, y^n, y^\star) = \mathbb{I}\{y^n \neq y^\star\}\,\mathbb{I}\{y^s = y^\star\}$ and $y^\star = loan = 0$:

$$\begin{aligned}
\mathrm{PCI}_{gender}(\text{Alice}) &= \mathbb{E}[ci(Y^s, Y^n, y^\star)] = \mathbb{P}(Y^n = 1,\, Y^s = 0) \\
&= \sum_u p(u)\, P_k^s(\{0\} \mid u)\, P_k^n(\{1\} \mid u) \\
&= 0.2 \times \tfrac{2}{3} \times 0.192 + 0.7 \times \tfrac{2}{3} \times 0.508 + 0.1 \times 0 \\
&\approx 0.026 + 0.237 = 0.263.
\end{aligned}$$

Table 2 gives the full results for Alice and Bob under these parameter choices: Pearl's PN/PS/PNS, PCI without witnesses ($\mathbf{W} = \emptyset$), and PCI with the *check-failed* witness active, with the latter further decomposed into its necessity and sufficiency marginals.

	Pearl			PCI	PCI ($\mathbf{W} = \{check\text{-}failed\}$)		
Person, Variable	PN	PS	PNS	($\mathbf{W} = \emptyset$)	Necessity	Sufficiency	Total
Alice (*gender*)	0.045	1.000	0.045	0.210	0.394	0.667	**0.263**
Alice (*credit*)	0.180	1.000	0.180	0.240	0.298	0.667	**0.199**
Bob (*gender*)	0.000	—	0.000	0.038	0.030	0.637	**0.019**
Bob (*credit*)	0.900	0.955	0.860	0.228	0.188	0.637	**0.119**

Table 2: Per-individual responsibility scores for *gender* and *credit* on $loan = 0$ in the stochastic OBCB model. Pearl's PN/PS/PNS, PCI without witnesses ($\mathbf{W} = \emptyset$), and PCI with the *check-failed* witness shown side by side; the rightmost three columns decompose PCI($\mathbf{W} = \{check\text{-}failed\}$) into its necessity and sufficiency marginals (their joint expectation is the bold total). Here the total happens to equal the product of the two marginals because the sufficiency measure $P_k^s(\cdot \mid \mathbf{u})$ is constant in $\mathbf{u}$ for both individuals; in general the joint $P_k^{s,n}$ does not factor into the product of its marginals. Bob's PS for gender is undefined because the conditioning event (female, bad, $loan = 1$) has probability zero in the OBCB model; the PCI marginals are well-defined throughout, since they integrate over $P_{\mathbf{U}}$ rather than conditioning on the factual outcome.

For Alice, PNS and PCI without witnesses both fail **D-A-rank** (p. 12): they rank *credit* above *gender* (0.180 vs. 0.045; 0.240 vs. 0.210). PCI with witnesses reverses this (0.263 vs. 0.199), satisfying **D-A-rank** as well as **D-A1** and **D-A2** (both values positive). Holding *check-failed* at its factual value blocks the credit-check bottleneck in noise realizations where Alice was checked, isolating gender's structural role. The *check-failed* witness is the decisive ingredient.

For Bob, **D-B2** and **D-B-rank** hold under all three methods: *credit* is ranked above *gender* throughout. **D-B1** requires strictly positive attribution for *gender*: PNS returns exactly 0.000 and therefore fails, because $P(loan = 1 \mid gender = \text{female}, credit = \text{bad}) = 0$ makes the counterfactual necessity calculation zero, missing gender's indirect role in opening the credit evaluation. PCI with witnesses returns 0.019, correctly capturing this indirect contribution. The cross-individual desideratum **D-comp** holds under both PCI ($0.263 > 0.019$) and PNS ($0.045 > 0.000$). A full verification of all seven desiderata is given at the end of this section.

The PNS and PCI (no witnesses) columns differ even though neither uses witnesses, because PCI still Γ-averages over candidate suspect sets. With $|\mathbf{S}| = 2$ PCI averages over $\{gender\}$, $\{credit\}$, and $\{gender, credit\}$ (each weighted $\frac{1}{3}$), whereas PNS evaluates a single counterfactual on one variable at a time. Property (i) below establishes that the two agree exactly when $|\mathbf{S}| = 1$ (single suspect, no witnesses, binary outcome), whereas the OBCB setup uses $|\mathbf{S}| = 2$ deliberately, to demonstrate joint-subset attribution.

3.1 Continuous Outcomes and the General Causal Impact Function

The causal impact function ci is a user-defined component of PCI, and different choices encode different causal questions. In the benchmarking experiments of Section 9, we instantiate ci using only the necessity component, since Halpern's actual causality has no sufficiency counterpart: necessity-only attribution allows a direct parallel comparison. The full framework incorporating both Y^s and Y^n is instantiated in the PNS binary example above and in the Absolute Difference example below; the OBCB analysis of this section (Tables 2–3) provides a concrete validation, since the PNS binary ci jointly conditions on $Y^s = y^\star$ and $Y^n \neq y^\star$ and it is precisely this combination that produces the correct **D-A-rank** ordering. Further experiments using the full ci on continuous outcomes are presented in the signal-with-mediation example (Section 4.3), the synthetic evaluation (Section 5), and the SIR benchmark (Section 10).

When the outcome variable Y is continuous rather than binary, indicators are replaced by a distance-based score that measures how far the necessity and sufficiency worlds land from the factual value $y^\star$. The natural choice, paralleling L_1-based scores such as ATE and SHAP, is the absolute difference.

Example 21 (Absolute Difference Impact Score)**.** *We can develop an analogous measure for continuous outcomes where we increase attribution with distance from the factual value $y^\star$ in the necessity world and decrease impact with distance to the factual value in the sufficiency world.*

$$ci(y^s, y^n, y^\star) = |y^n - y^\star| - |y^s - y^\star|.$$

With the full machinery in place — structural model, variable selection, alternative value distribution, joint necessity-sufficiency measure, and causal impact function — PCI delivers the following properties.

(i) **Generalisation of PN/PS/PNS.** When $\mathbf{S} = \{X_k\}$, $\mathbf{W} = \emptyset$, and $\mathrm{dom}\,(X_k) = \{0, 1\}$, the PNS binary ci recovers Pearl's probability of necessity and sufficiency exactly; PN and PS are recovered by the corresponding one-sided ci choices.

(ii) **Connection to actual causality.** For suitable Γ and Δ, PCI recovers Halpern's actual causality judgements (Section 7).

(iii) **Necessity–sufficiency decomposition.** P^s_k and P^n_k are defined and interpretable independently; practitioners may choose a ci that uses either alone or both, depending on whether the causal question concerns necessity, sufficiency, or both.

(iv) **Witness-mediated symmetry breaking.** Holding intermediate variables fixed via $\mathbf{W}$ disambiguates overdetermination and undercutting scenarios where necessity-only methods assign equal or zero attribution, as demonstrated above for Alice and Bob, and further developed in Section 4 (SHAP and Causal SHAP comparison), Section 5 (synthetic overdetermination and undercutting archetypes), and Section 9 (witness ablation on the scaled throwing problem).

(v) **Composability.** Γ, Δ, and ci are probability distributions and a measurable function respectively; they slot naturally into Bayesian pipelines, accept posteriors as inputs, and admit Monte Carlo estimation without modification to the core formalism.

3.2 Back to the Running Example: Verifying the Desiderata

Table 3 verifies the seven desiderata introduced at the start of this section (p. 12) against the values in Table 2. PCI with witnesses satisfies all seven; PCI without witnesses fails **D-A-rank**; PNS fails **D-A-rank** and **D-B1**. (The table uses the shorthand $|R(V \mid \text{person})|$ for $|R(V \rightsquigarrow \mathit{loan} \mid \text{person})|$.)

Desideratum	PNS	PCI (no wit.)	PCI (with wit.)
D-A1: $\lvert R(\mathit{gender} \mid \text{Alice})\rvert > 0$	✓ 0.045	✓ 0.210	✓ 0.263
D-A2: $\lvert R(\mathit{credit} \mid \text{Alice})\rvert > 0$	✓ 0.180	✓ 0.240	✓ 0.199
D-A-rank: $\lvert R(\mathit{gender} \mid \text{Alice})\rvert > \lvert R(\mathit{credit} \mid \text{Alice})\rvert$	× 0.045<0.180	× 0.210<0.240	✓ 0.263>0.199
D-B1: $\lvert R(\mathit{gender} \mid \text{Bob})\rvert > 0$	× 0.000	✓ 0.038	✓ 0.019
D-B2: $\lvert R(\mathit{credit} \mid \text{Bob})\rvert > 0$	✓ 0.860	✓ 0.228	✓ 0.119
D-B-rank: $\lvert R(\mathit{credit} \mid \text{Bob})\rvert > \lvert R(\mathit{gender} \mid \text{Bob})\rvert$	✓ 0.860>0.000	✓ 0.228>0.038	✓ 0.119>0.019
D-comp: $\lvert R(\mathit{gender} \mid \text{Alice})\rvert > \lvert R(\mathit{gender} \mid \text{Bob})\rvert$	✓ 0.045>0.000	✓ 0.210>0.038	✓ 0.263>0.019

Table 3: Verification of the seven desiderata (p. 12) against the values in Table 2. PCI with witnesses satisfies all seven; PCI without witnesses fails **D-A-rank**; PNS fails **D-A-rank** and **D-B1**.

*PNS fails **D-A-rank** and **D-B1**. The D-A-rank failure reflects that without a* `check-failed` *witness, necessity-based attribution cannot distinguish Alice's gender (which blocked the evaluation entirely) from her credit (which would only have mattered had she been evaluated). The D-B1 failure reflects* *that PNS assigns Bob's gender exactly zero: because $P(loan = 1 \mid female, bad) = 0$, the counterfactual necessity calculation returns zero, missing gender's indirect role in opening the credit evaluation. The $check\text{-}failed$ witness resolves both.*

To close this section, it is worth comparing PCI's decomposition with the one already present in Pearl's framework. Pearl's framework defines three related but distinct binary scores: the probability of necessity $\mathrm{PN} = P(Y_{x'} = y' \mid X = x^*, Y = y^*)$, the probability of sufficiency $\mathrm{PS} = P(Y_{x^*} = y^* \mid X = x', Y = y')$, and their joint $\mathrm{PNS} = P(Y_{x^*} = y^*, Y_{x'} = y')$. The first two are conditional probabilities; PNS is unconditional, and the three satisfy $\mathrm{PNS} = \mathrm{PN} \cdot P(Y_{x^*} = y^*) = \mathrm{PS} \cdot P(Y_{x'} = y')$ rather than $\mathrm{PNS} = \mathrm{PN} \cdot \mathrm{PS}$ (which would require $Y_{x^*} \perp Y_{x'}$, an assumption typically violated since the two potential outcomes share exogenous noise). PCI admits a related split: with $ci = \mathbb{I}\{y^n \neq y^\star\}\, \mathbb{I}\{y^s = y^\star\}$,

$$\mathbb{E}[ci(Y^s, Y^n, y^\star)] = \int P_k^s(\{y^\star\} \mid \mathbf{u})\, P_k^n(\{1-y^\star\} \mid \mathbf{u})\, dP_{\mathbf{U}},$$

giving a *sufficiency marginal* and a *necessity marginal* that record how often, on average, restoring the factual values of a suspect set containing X_k reproduces the factual outcome, and how often substituting alternatives overturns it. The two marginals are independent given $\mathbf{u}$.

The components are not identical to Pearl's: PN and PS condition on the factual outcome and consider a single counterfactual flip; PCI's marginals integrate over $P_{\mathbf{U}}$, average over suspect sets weighted by Γ, and hold the witness configurations fixed when active. Table 2 (presented earlier) places both decompositions side by side for $gender$ and $credit$.

For Alice's gender, $\mathrm{PN} = 0.045$ is the bare probability that a male with bad credit would have been approved; PCI's necessity marginal is 0.394, nearly an order of magnitude higher, because the $check\text{-}failed$ witness keeps the credit-check bottleneck active when the male counterfactual would otherwise have been checked and might still have failed, structural information that the unconditional PN cannot register. For Bob's gender, PN is exactly 0 (a female with bad credit cannot be approved in this model) and PS is undefined; PCI's marginals remain well-defined at 0.030 and 0.637, recovering the indirect role of gender in opening the evaluation.

The next section turns from comparing PCI with Pearl's family of counterfactual scores to comparing it with the SHAP family, methods grounded in cooperative game theory rather than in structural counterfactual reasoning. Section 4 re-runs the OBCB analysis under plain and Causal SHAP, then introduces a continuous mediation example where the two SHAP variants diverge from each other and from PCI, with explicit desiderata recorded throughout.

4 Examples and Comparison to SHAP and Causal SHAP

Where we are. Section 3 defined PCI and verified, on Alice and Bob, that the resulting attribution satisfies all seven desiderata of that section, where PN/PS/PNS fail at least two. The natural next question is how PCI compares to the dominant practical alternative for feature attribution in the ML literature, SHAP and Causal SHAP, since these are the tools practitioners typically reach for. This section runs that comparison on two examples, recording where each method passes or fails an explicit desideratum. We first define SHAP and Causal SHAP formally (Section 4.1), then re-run OBCB (Section 4.2), where the two SHAP variants coincide because the feature set is exogenous, so the divergence between PCI and SHAP is the only contrast to draw. We then introduce a continuous mediation example (Section 4.3) where plain and Causal SHAP do diverge (one desideratum, D-YX, is fixed by the interventional correction; others, D-YM, D-MX, D-MXY, are not), and tabulate the desiderata in Section 4.4. Section 4.5 traces every failure to one of three structural gaps in the SHAP framework that PCI's witness mechanism, suspect-set distribution Γ_s, and realised-outcome reference jointly close.

Shapley values have become one of the most widely used tools for explaining the predictions of complex machine learning models. Grounded in cooperative game theory, they provide a principled decomposition of a model's prediction into additive contributions from individual features. Janzing et al. [2020] argued that the conditional expectation in plain SHAP is the conceptually wrong way to model feature removal: a dropped feature should be marginalised under a do-intervention, not under conditioning. Heskes et al. [2020] propose Causal SHAP, which implements this prescription by replacing the observational marginal used for features outside the active coalition with an interventional distribution: when a dropped feature is causally downstream of a coalition member, its distribution under the intervention differs from its marginal, and Causal SHAP corrects for this using Pearl's do-calculus.[14] Formal definitions of both are given in Section 4.1.

Despite this repair, both methods share deeper limitations. First, both plain and Causal SHAP can assign non-zero responsibility to variables that play no causal role with respect to the target: attribution can run against the direction of the causal graph, a failure the interventional correction does not address. Second, both methods explain a model's predicted output averaged over the population, not the individual factual outcome of a specific person: when the goal is *causal responsibility attribution*, this produces further failures that are not always made explicit in the literature. The first limitation will surface in our signal example as violations of D-YX, D-YM, and D-MX; the second will surface in OBCB as the D-A-rank overdetermination effect, and in the signal example as Instance 2's budget collapse when the prediction equals its baseline.

We trace these failures through two examples, comparing plain SHAP, Causal SHAP, and PCI against the desiderata of Section 3 and some further desiderata related to the example we will employ. We first return to the OBCB model: it permits exact analytic computation and the desiderata are already known. In OBCB the feature set contains only exogenous roots, so plain and Causal SHAP coincide: the comparison is short. We then turn to a continuous mediation example where downstream variables are observable model inputs; there the two methods diverge, and we examine what Causal SHAP fixes and what it does not.

4.1 SHAP and Causal SHAP

In this section $\mathcal{S} \subseteq N$ denotes a SHAP coalition (a subset of features held to their factual values during evaluation), playing the same role here that $\mathbf{C} \subseteq \mathbf{S}$ played in Section 3. We use the calligraphic $\mathcal{S}$ to keep the SHAP coalition typographically distinct from PCI's suspect set $\mathbf{S}$ on the printed page; the SHAP literature's plain S is the same object.

Definition 22 (SHAP)**.** *A cooperative game is a pair (N, v), where $N = \{1, \ldots, n\}$ is the set of players and $v : 2^N \to \mathbb{R}$ is the* characteristic function *mapping each subset (coalition) $\mathcal{S} \subseteq N$ to the value that coalition can produce. The Shapley value [Shapley, 1953] assigns each player i a payoff ϕ_i, its share of the total value $v(N) - v(\emptyset)$, and is the unique attribution satisfying four axioms:*

1. ***Efficiency**: $\sum_i \phi_i = v(N) - v(\emptyset)$*
2. ***Symmetry**: If $v(\mathcal{S} \cup \{i\}) = v(\mathcal{S} \cup \{j\})$ for all $\mathcal{S}$, then $\phi_i = \phi_j$*
3. ***Dummy**: If $v(\mathcal{S} \cup \{i\}) = v(\mathcal{S})$ for all $\mathcal{S}$, then $\phi_i = 0$*
4. ***Linearity**: For two games v_1, v_2: $\phi_i(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 \phi_i(v_1) + \alpha_2 \phi_i(v_2)$*

The unique solution is:

$$\phi_i = \sum_{\mathcal{S} \subseteq N \setminus \{i\}} \frac{|\mathcal{S}|!\,(|N| - |\mathcal{S}| - 1)!}{|N|!} \left[v(\mathcal{S} \cup \{i\}) - v(\mathcal{S}) \right] \tag{5}$$

The weight $\frac{|\mathcal{S}|!(|N|-|\mathcal{S}|-1)!}{|N|!}$ equals the probability that, in a uniformly random ordering of all players, exactly the members of $\mathcal{S}$ precede i. Equivalently, ϕ_i is the average marginal contribution of i across all orderings. Lundberg and Lee [2017] apply Shapley values to explain model predictions by taking players to be input features, and the characteristic function to be the expected model output given a subset of features:

$$v(\mathcal{S}) = \mathbb{E}_{X_{\bar{\mathcal{S}}}} \left[f(X_{\mathcal{S}} = x_{\mathcal{S}},\, X_{\bar{\mathcal{S}}}) \right] \tag{6}$$

[14] A different counterfactual line is taken by Sharma et al. [2022], whose CF-Shapley values use a structural causal model to attribute the change in a metric from a single fixed reference x^{ref} to the observed x: $\phi_j \propto \sum_{\mathcal{S} \subseteq N \setminus \{j\}} [f(x_{\mathcal{S}}, x_j^{\text{ref}}) - f(x_{\mathcal{S}}, x_j)]$. This is the closest neighbour to PCI in spirit (single contrastive scenario, *do*-style intervention) and differs in three ways. *(i)* CF-Shapley commits to one reference x^{ref}, while PCI integrates over an alternative-value distribution $\Delta(\mathbf{x})$ and need not pick a single contrast. *(ii)* CF-Shapley returns a one-axis contrast $f(\cdot, x_j^{\text{ref}}) - f(\cdot, x_j)$, while PCI reports a joint (Y^s, Y^n) measure and applies a user-chosen ci, decomposing necessity and sufficiency separately. *(iii)* CF-Shapley has no witness mechanism, so it cannot break the overdetermination and undercutting symmetries that PCI's witness set $\mathbf{W}$ resolves on the OBCB and signal walkthroughs of this section.

where $\bar{\mathcal{S}} = N \setminus \mathcal{S}$ and missing features are marginalized over their marginal distribution.

Definition 23 (Causal SHAP)**.** *Causal SHAP replaces the characteristic function with an interventional expectation [Heskes et al., 2020]:*

$$v_{\mathit{causal}}(\mathcal{S}) = \mathbb{E}\big[f(X) \mid do(X_{\mathcal{S}} = x_{\mathcal{S}})\big] = \int P(X_{\bar{\mathcal{S}}} \mid do(X_{\mathcal{S}} = x_{\mathcal{S}}))\, f(X_{\bar{\mathcal{S}}}, x_{\mathcal{S}})\, dX_{\bar{\mathcal{S}}} \tag{7}$$

By the rules of do-calculus, intervening on $X_{\mathcal{S}}$ only affects the distribution of descendants of $X_{\mathcal{S}}$. For non-descendants, $P(X_j \mid do(X_{\mathcal{S}} = x_{\mathcal{S}})) = P(X_j)$. Writing $X_{\bar{\mathcal{S}},\,\mathrm{nd}}$ for the features in $\bar{\mathcal{S}}$ that are not descendants of any member of $\mathcal{S}$, and $X_{\bar{\mathcal{S}},\,\mathrm{d}}$ for those that are, equation (7) is equivalent to:

$$v_{\mathit{causal}}(\mathcal{S}) = \mathbb{E}_{X_{\bar{\mathcal{S}},\,\mathrm{nd}} \sim P(\cdot),\; X_{\bar{\mathcal{S}},\,\mathrm{d}} \sim P(\cdot \mid do(X_{\mathcal{S}} = x_{\mathcal{S}}))}\big[f(X_{\bar{\mathcal{S}}}, x_{\mathcal{S}})\big] \tag{8}$$

The Shapley formula (5) is otherwise unchanged. The method retains all four Shapley axioms.

At the level of subset weighting alone, PCI (with $J = 1$, $K = |\mathbf{S}|$) and SHAP are not distinguishable: Γ's subset weights match the Shapley kernel. The distinction lies in the integrand. SHAP averages marginal contributions using the observational conditional $\mathbb{E}[f(\mathbf{X}_{\mathcal{S}} = \mathbf{x}_{\mathcal{S}}, \mathbf{X}_{\bar{\mathcal{S}}})]$, marginalising over the empirical distribution of non-coalition features; a feature merely correlated with a true cause can therefore receive positive attribution. PCI evaluates counterfactual outcomes $P(Y \in A \mid \mathbf{u}, \mathrm{do}(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{t}^{\star}))$ computed via the structural model, so attribution flows only along genuine causal paths. The witness mechanism adds a further distinction: by holding intermediate variables fixed, PCI can resolve overdetermination and undercutting scenarios where SHAP assigns equal or incorrect attribution. The remainder of this section makes both points concrete on two examples.

4.2 OBCB: SHAP and Causal SHAP

To apply the SHAP and Causal SHAP definitions of Section 4.1 to OBCB we must specify both a feature set N and a function f on the joint feature space; the value functions and Shapley sums are derived from these. The natural choice for f is the conditional expectation of the outcome, $f(x_N) = \mathbb{E}[\mathit{loan} \mid X_N = x_N]$. But the OBCB PSCM has five variables — gender, credit, check, $\mathit{check\text{-}failed}$, $\mathit{loan\_if\_checked}$ — so the choice of N is non-trivial. Two options are natural:

- **Option A: two upstream roots.** $N = \{\mathit{gender}, \mathit{credit}\}$. Both features are exogenous; the three internal variables (check, $\mathit{check\text{-}failed}$, $\mathit{loan\_if\_checked}$) are integrated out inside f. This is the smallest sufficient feature set and matches the way the model is normally described: predict approval from gender and credit.
- **Option B: two roots plus the mediator.** $N' = \{\mathit{gender}, \mathit{credit}, \mathit{check\text{-}failed}\}$. The mediator becomes an observable model input, the same status it has in the signal-with-mediation example (§4.3). This is the comparison PCI's witness mechanism naturally invites: PCI accesses $\mathit{check\text{-}failed}$ as a witness, so allowing SHAP to access it as a feature is the fair foil.

We work through both. Option A is simpler and exposes the qualitative failures of plain SHAP. Option B then asks whether admitting the mediator rescues Causal SHAP.

Option A: two upstream roots. Here $f(g, c) = P(\mathit{loan} = 1 \mid g, c)$ takes values $f(\mathrm{F}, \mathrm{bad}) = 0.000$, $f(\mathrm{F}, \mathrm{good}) = 0.180$, $f(\mathrm{M}, \mathrm{bad}) = 0.045$, $f(\mathrm{M}, \mathrm{good}) = 0.900$, with the three internal variables integrated out. We apply SHAP to the rejection probability $g(x) = 1 - f(x)$ for Alice and Bob from Example 6, attributing responsibility for the denial outcome.[15] By the SHAP definition (and equivalently by Definition 23 once we note that no dropped feature is a descendant of any coalition member), the value function on the 2-feature game and its closed-form specialisation under independent uniform marginals are derived in Appendix A (§A.3). Plugging in the four values of f listed at the start of Option A yields Table 4; pushing those through the closed-form Shapley gives the magnitudes in Table 5, which compares against PNS and PCI.

[15] Because $g = 1 - f$, the characteristic functions satisfy $v_g(\mathcal{S}) = 1 - v_f(\mathcal{S})$ and $\phi_i^g = -\phi_i^f$; we compute ϕ^f and flip signs. With $|N| = 2$ and uniform marginals $P(\mathit{gender}) = P(\mathit{credit}) = \frac{1}{2}$, equation (5) reduces to $\phi_i = \frac{1}{2}[v(\{i\}) - v(\emptyset)] + \frac{1}{2}[v(N) - v(N \setminus \{i\})]$.

$\mathcal{S}$	$v(\mathcal{S})$ for Alice	$v(\mathcal{S})$ for Bob
$\emptyset$	0.281	0.281
$\{gender\}$	0.090	0.473
$\{credit\}$	0.023	0.023
$\{gender, credit\}$	0.000	0.045

Table 4: Coalition values $v(\mathcal{S}) = \mathbb{E}[f(X) \mid X_{\mathcal{S}} = x^{\star}_{\mathcal{S}}]$ for the 2-feature game ($N = \{gender, credit\}$). $v(\emptyset) = \mathbb{E}[f(X)] = 0.281$ is the population baseline; $v(N)$ is the factual prediction $f(x^{\star})$. Plain SHAP and Causal SHAP coincide on this game (neither feature is downstream of the other), so a single column suffices per individual.

Person	Feature	PNS	SHAP	ATE	CATE	ITE	PCI (no wit.)	PCI (with wit.)
Alice	*gender*	0.045	0.107	0.383	0.045	0.045	0.210	**0.263**
Alice	*credit*	0.180	0.174	0.518	0.180	0.180	0.240	**0.199**
Bob	*gender*	0.000	-0.107	0.383	0.045	0.000	0.038	**0.019**
Bob	*credit*	0.860	0.343	0.518	0.855	0.895	0.228	**0.119**

Table 5: Responsibility attributions for Alice and Bob. SHAP values are signed (ϕ_i^g for $P(loan = 0 \mid x)$); desiderata compare magnitudes $|\cdot|$. ATE is a population quantity (identical across the two persons by feature); CATE conditions on the *other* covariate (credit for the gender rows; the person's gender for the credit rows); ITE abducts the individual's exogenous noise from the full factual record (covariates *and* the observed denial) and takes the counterfactual contrast on that pinned noise. PNS and PCI values from Table 2; ATE/CATE/ITE computed by direct PSCM evaluation in `docs/source/obcb_computations.ipynb`.

ATE and CATE fail the same two individual-level desiderata that PNS fails, and for related reasons. ATE attributes responsibility at the population level only, so its values are identical across Alice and Bob, immediately violating **D-comp** ($|R(gender \mid \text{Alice})| = |R(gender \mid \text{Bob})| = 0.383$). CATE conditions on observable covariates but inherits PNS's blind spot via the *check-failed* propagation discussed in Section 2: for Alice it returns 0.045 for gender and 0.180 for credit, the same ranking as PNS_c and the same **D-A-rank** failure. CATE also fails **D-comp**, since $\text{CATE}(gender \mid credit{=}\text{bad}) = 0.045$ is the same number for Alice and Bob.

The individual treatment effect (ITE) is the natural endpoint of this ladder: where ATE marginalises all exogenous noise and CATE conditions only on observed covariates, the ITE abducts the unit's noise from its *full* factual record — covariates *and* the observed denial — and then takes the counterfactual contrast holding that abducted noise fixed, the abduction–action–prediction reading already used for PN in Section 2. This buys back exactly the information CATE discards. The ITE separates the two applicants on **D-comp** where CATE cannot: $\text{ITE}(gender \mid \text{Alice}) = 0.045$ but $\text{ITE}(gender \mid \text{Bob}) = 0$, because abduction places Bob in the checked-and-failed branch, where being male is not what blocked him — flipping him to female (still bad credit) leaves him denied with certainty. The ITE is also a genuine counterfactual rather than an intervention: because it conditions on the outcome, $\text{ITE}(credit \mid \text{Bob}) = 0.895$ rather than the interventional $P(loan{=}1 \mid do(credit{=}\text{good}), gender{=}\text{male}) = 0.90$, as conditioning on Bob's denial shifts posterior mass toward the unchecked branch. But the ITE does *not* fix **D-A-rank**: for Alice it still returns 0.180 for credit against 0.045 for gender, the same inversion as PNS_c and CATE. Abduction pins the *noise* but not the *mechanism* — the counterfactual still routes through a hypothetical credit check in the 20% of Alice's posterior in which she was checked, so credit retains spurious responsibility. Recovering the correct ranking requires PCI's witness mechanism, which pins *check-failed* at its factual value (dominantly the unchecked branch for Alice, where credit is causally disconnected from the loan) rather than merely integrating over its noise. In this sense PCI is the ITE *plus* the structural machinery needed to see the mediated path: it inherits the ITE's individual-level, counterfactual character and adds the context-fixing the ITE alone lacks.

SHAP fails two desiderata. The primary failure is **D-A-rank**: Alice's credit outranks her gender ($0.174 > 0.107$). This is an overdetermination effect: *credit*=bad correlates with the *check-failed* mechanism that blocked Alice's credit evaluation, so SHAP's observational marginalization inflates credit attribution above gender's. PCI includes the factual $check\text{-}failed$ value directly, isolating gender's upstream causal role and recovering the correct ranking. PNS fails **D-A-rank** for the same reason; it also fails **D-B1** by returning exactly zero for Bob's gender (see Table 3).

The second SHAP failure is **D-comp**: $|\phi^g_{gender}| = 0.107$ for both Alice and Bob. But gender played a strictly larger causal role in Alice's rejection than in Bob's. For Alice, $gender$=female was sufficient on its own to produce rejection: it blocked the credit evaluation entirely, so her credit was never a factor. For Bob, $gender$=male was a facilitating condition: it triggered the credit check, but $credit$=bad was the actual driver of the denial. SHAP cannot distinguish these because it measures how far each feature value deviates from the population prediction mean: female and male are equidistant deviations in opposite directions, so they receive equal magnitude regardless of their individual causal weight. This is an instance of a deeper limitation: SHAP explains predicted outputs, not individual factual outcomes, which we examine further in Section 4.3.

Causal SHAP coincides exactly with plain SHAP under Option A: both features are exogenous roots, so $P(credit \mid do(gender = g)) = P(credit)$ and vice versa, no dropped feature is a descendant of any coalition member, and $v_{\text{causal}}(\mathcal{S}) = v_{\text{plain}}(\mathcal{S})$ for every coalition. Option A therefore reduces to a single SHAP analysis, and the failures above belong to both methods.

Option B: two roots plus the mediator. Throughout this subsection we abbreviate $gender$, $credit$, and $check\text{-}failed$ as g, c, and cf in mathematical expressions and table cells; running prose retains the full names. Now we admit $check\text{-}failed$ as a third model input and apply both methods to the 3-feature game $N' = \{gender, credit, check\text{-}failed\}$. This requires defining f on $\{0,1\}^3$ rather than on $\{0,1\}^2$. The natural observational extension is again the conditional expectation $\tilde{f}(g, c, cf) = \mathbb{E}[loan \mid gender = g,\ credit = c,\ check\text{-}failed = cf]$. This is well defined for three of the four (c, cf) combinations, but not the fourth: a failed check requires that a check happened *and* the credit was bad, so $check\text{-}failed = 1$ together with $credit =$ good is impossible under the PSCM and has no observational data to condition on. We leave $\tilde{f}$ undefined on this cell.

Both methods need a value function $v(\mathcal{S})$: the expected value of $\tilde{f}$ when the features in $\mathcal{S}$ are held at their factual values $x^\star_{\mathcal{S}}$ and the remaining features are averaged out. The methods differ in how that average is taken. Plain SHAP averages over each dropped feature using its own observational marginal $P(X_i)$, with the dropped features treated as independent of one another. Causal SHAP averages over the joint distribution the PSCM produces under the intervention $do(X_{\mathcal{S}} = x^\star_{\mathcal{S}})$, so the dropped features keep whatever joint dependencies the PSCM imposes on them. Whether either average ever lands on the impossible cell (c=1, cf=1) (and so whether $v(\mathcal{S})$ is finite) depends on both the method and on which features are held fixed at which factual values. Table 6 works out all eight coalition values for both individuals.

	Alice (g, c, cf)=$(0,0,0)$		Bob (g, c, cf)=$(1,0,1)$	
$\mathcal{S}$	v_{plain}	v_{causal}	v_{plain}	v_{causal}
$\emptyset$	**NaN**	0.281	**NaN**	0.281
$\{g\}$	**NaN**	0.090	**NaN**	0.473
$\{c\}$	0.007	0.023	0.007	0.023
$\{cf\}$	0.270	0.270	**NaN**	**NaN**
$\{g, c\}$	0.000	0.000	0.014	0.045
$\{g, cf\}$	0.090	0.090	**NaN**	**NaN**
$\{c, cf\}$	0.000	0.000	0.025	0.025
N'	0.000	0.000	0.050	0.050

Table 6: Coalition values for the 3-feature game $N' = \{gender, credit, check\text{-}failed\}$. Bold **NaN** entries flag coalitions whose value function evaluates $\tilde{f}$ at the impossible cell (c=1, cf=1) with positive weight. Plain SHAP and Causal SHAP coincide whenever neither $credit$ nor $check\text{-}failed$ is dropped, since the only structural coupling in the model is between these two; they diverge whenever one is in $\mathcal{S}$ and the other is not.

Plain SHAP. Plain SHAP treats $credit$ and $check\text{-}failed$ as independent, so whenever both are dropped from the coalition the average runs over all four (c, cf) combinations, with the impossible (c=1, cf=1) combination receiving weight $P(c{=}1) \cdot P(cf{=}1) = 0.5 \cdot 0.275 \approx 0.140$, the same nonzero weight as any other corner. For Alice, both $credit$ and $check\text{-}failed$ are dropped at $\mathcal{S} \in$

$\{\emptyset, \{gender\}\}$, the first two NaN entries in Table 6. For Bob the same two coalitions fail, and two more break for an additional reason: at $\mathcal{S} = \{check\text{-}failed\}$ and $\mathcal{S} = \{gender, check\text{-}failed\}$ the value $cf^\star = 1$ is held fixed while *credit* is sampled from its marginal, so the c=1 branch enters with weight 0.5 and again hits the impossible cell. Plain SHAP gives undefined Shapley values for both individuals.[16]

Causal SHAP for Alice. Causal SHAP samples cf following its structural equation $cf = check \cdot (1 - c)$, with $check \sim \mathrm{Bern}(p_{\text{check}}[g])$, given whatever (g, c) values appear in the integrand. When $c = 1$ the equation forces $cf = 0$ regardless of *check*, so the $(c{=}1, cf{=}1)$ combination is never produced. Every Alice entry in the v_{causal} column of Table 6 is therefore finite. Plugging Alice's column into the Shapley formula gives magnitudes $|\phi_{gender}| = 0.097$, $|\phi_{credit}| = 0.176$, $|\phi_{check\text{-}failed}| = 0.007$, so $|\phi_{credit}| > |\phi_{gender}|$ and **D-A-rank** fails: credit outranks gender, exactly the same pathology plain SHAP exhibited on the 2-feature game.

Causal SHAP for Bob. Two coalitions break for Bob: $\mathcal{S} = \{check\text{-}failed\}$ and $\mathcal{S} = \{gender, check\text{-}failed\}$. Both hold *check-failed* at Bob's factual value $cf^\star = 1$ and average *credit* over its marginal. To see why this is fatal, apply Definition 23 to the first:

$$v_{\text{causal}}(\{cf\}) \;=\; \mathbb{E}\Big[\tilde{f}(X) \,\Big|\, do(cf = 1)\Big] \;=\; \tfrac{1}{4} \sum_{g \in \{0,1\}} \sum_{c \in \{0,1\}} \tilde{f}(g, c, 1).$$

The uniform $\frac{1}{4}$ weights come from *gender* and *credit* sitting upstream of *check-failed* in the causal graph: intervening on *check-failed* overrides the structural equation that would normally determine it from credit and check, but does not alter the marginal distributions of the upstream features. Two of the four summands evaluate $\tilde{f}$ at $(g, c{=}1, cf{=}1)$ — the impossible cell — with weight $\frac{1}{4}$ each. So $v_{\text{causal}}(\{cf\})$ is undefined, and so is $v_{\text{causal}}(\{g, cf\})$ for the same reason. These two coalitions appear in the Shapley sum for every feature, so all three of Bob's Causal SHAP values are NaN (Bob's columns in Table 6).

Why Bob breaks and Alice does not. The same two coalitions arise for Alice; the difference is the value of $cf^\star$ being held fixed. With Alice's $cf^\star = 0$, the c=1 summand in the formula above evaluates $\tilde{f}(g, 1, 0) = p_{\text{check}}[g] \cdot \text{loan\_prob}[g, 0]$, which is defined because the PSCM does allow $(c{=}1, cf{=}0)$: when credit is good, cf is forced to 0 deterministically, so the conditional expectation has a value. With Bob's $cf^\star = 1$, the c=1 summand evaluates $\tilde{f}(g, 1, 1)$, the impossible cell, "good credit and a failed check", which the PSCM rules out, since failing the credit check requires bad credit by the structural equation $cf = check \cdot (1 - c)$.

The mechanism producing this asymmetry is the intervention itself. When the PSCM runs forward, *credit* and *check-failed* are linked by $cf = check \cdot (1 - c)$, which keeps c=1 and cf=1 from co-occurring. When Causal SHAP intervenes via $do(cf = cf^\star)$, this link is overridden: *check-failed* is fixed externally and *credit* is sampled freely. The override always produces a $(c{=}1, cf^\star)$ combination with positive weight; whether $\tilde{f}$ has a value for that combination depends on which $cf^\star$ value Causal SHAP is intervening on.

Promoting the mediator to a feature therefore does not rescue Causal SHAP: it fails the desideratum on Alice and breaks down outright on Bob.

The OBCB analysis exposed two structural failure modes: SHAP's overdetermination effect on the 2-feature game, and Causal SHAP's breakdown once the mediator is admitted as a feature under Option B. The second of these depends on a structural feature of the OBCB PSCM: the equation $cf = check \cdot (1 - c)$ ties *credit* and *check-failed* together, so Option B's natural feature set contains one combination of input values ("good credit and a failed check") that the model is not defined on. To see whether the same families of failure appear without this complication, we now turn to the signal-with-mediation chain $X \to M \to Y$, where the mediator is naturally an observable input, $\tilde{f}$ is defined everywhere, and no impossible-cell issue arises. Causal SHAP's failures there are not artifacts of an undefined $\tilde{f}$ but properties of the method itself.

[16] The underlying issue is that plain SHAP does not see that *credit* and *check-failed* are linked by the structural equation $cf = check \cdot (1 - c)$, and so cheerfully samples a combination the PSCM rules out.

4.3 Signal with Mediation

Example 24 (Signal with mediation)**.** *We work with the following* ***generative model****: X sends a signal, M mediates it with some noise, and Y is the noisy final outcome:*

$$X \sim \mathcal{N}(0.5,\ 0.25) \tag{9}$$

$$M = X + \varepsilon_M, \quad \varepsilon_M \sim \mathcal{N}(0,\ 0.1) \tag{10}$$

$$Y = M + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0,\ 0.1) \tag{11}$$

The causal graph is $X \to M \to Y$, a simple mediation chain. All noise terms are independent of each other and of X. Since everything is jointly Gaussian, all conditional expectations are linear functions of the conditioning variables, and no approximations are required.

Let $R(X \rightsquigarrow Y)$ denote the causal responsibility of X for Y. Following the convention of Section 3, the desiderata are stated in terms of absolute magnitudes $|R(\cdot)|$, allowing for directionality. The ***qualitative causal responsibility desiderata*** *for this example are:*

$$\textbf{\textit{D-XY}} \quad |R(X \rightsquigarrow Y)| > 0$$

$$\textbf{\textit{D-MY}} \quad |R(M \rightsquigarrow Y)| > 0$$

$$\textbf{\textit{D-XM}} \quad |R(X \rightsquigarrow M)| > 0$$

$$\textbf{\textit{D-YX}} \quad |R(Y \rightsquigarrow X)| = 0$$

$$\textbf{\textit{D-YM}} \quad |R(Y \rightsquigarrow M)| = 0$$

$$\textbf{\textit{D-MX}} \quad |R(M \rightsquigarrow X)| = 0$$

A more subtle desideratum concerns telling apart the roles *of the two genuine causes. X reaches Y only through the mediator M, whereas M acts on Y directly; a good attribution method should be able to distinguish these two roles rather than collapse them into the same number. One systematic way to register the distinction is to require that the two causes receive different, rankable scores, and we adopt the convention of assigning the larger score to the cause closer to the outcome:*

$$\textbf{\textit{D-MXY}} \quad |R(M \rightsquigarrow Y)| > |R(X \rightsquigarrow Y)|$$

The direction of the inequality is a convention rather than a law of responsibility; what matters is that the method separates the direct from the indirect path at all, and scoring the closer cause higher is one principled, consistent way to register that separation. We apply this convention throughout.

The key ***distributional facts*** *we will use are:*

$$\mathbb{E}[X] = \mathbb{E}[M] = \mathbb{E}[Y] = 0.5, \tag{12}$$

$$\big(\mathit{Var}(X),\ \mathit{Var}(M),\ \mathit{Var}(Y)\big) = (0.25,\ 0.35,\ 0.45), \tag{13}$$

$$\big(\mathit{Cov}(X,M),\ \mathit{Cov}(X,Y),\ \mathit{Cov}(M,Y)\big) = (0.25,\ 0.25,\ 0.35). \tag{14}$$

For each target variable, the optimal predictor under squared loss is the conditional expectation, computable exactly in this Gaussian model:

$$f_Y(X,M) = \mathbb{E}[Y \mid X,M] = M \quad (\{X,M\} \Rightarrow Y) \tag{15}$$

$$f_M(X,Y) = \mathbb{E}[M \mid X,Y] = 0.5X + 0.5Y \quad (\{X,Y\} \Rightarrow M) \tag{16}$$

$$f_X(M,Y) = \mathbb{E}[X \mid M,Y] = 0.5 + \frac{5}{7}(M - 0.5) \quad (\{M,Y\} \Rightarrow X) \tag{17}$$

The distributional facts and the three conditional expectations above are derived in Appendix B (§B.1–§B.2).

4.4 Computations and Desiderata Analysis

We analyze two instances. **Instance 1** sets $X^\star = M^\star = Y^\star = 1$: every variable lies above its mean, the prediction is non-trivial at every target, and SHAP, Causal SHAP, and PCI all return non-zero values. **Instance 2** sets $X^\star = M^\star = Y^\star = 0.5$: every variable sits at its mean, the prediction equals its baseline, and SHAP's attribution budget $f(x^\star) - \mathbb{E}[f(X)]$ collapses to zero. We work through Instance 1 first, then return to Instance 2 to isolate what each method is really tracking.

Computing the three methods. At Instance 1 ($X^\star = M^\star = Y^\star = 1$) all three methods admit closed-form evaluation in this Gaussian model. We defer the full per-target coalition-value and Shapley computations for plain and Causal SHAP, and the per-$(\mathbf{C}, \mathbf{T})$ PCI derivation, to Appendix B (§B.3–§B.5); the resulting attributions populate Table 7. One computed fact drives the D-YX discussion below and is worth stating here.

Plain SHAP assigns $\phi_Y = 0.139 > 0$ for target X, even though Y does not cause X. This arises because $v(\{Y\}) = 0.778 > 0.5$: observing $Y{=}1$ is statistically informative about $X{=}1$ through the chain $X \to M \to Y$. SHAP cannot distinguish "Y is informative about X" from "Y causes X" because the value function is grounded in observational conditionals, which are symmetric under reversal of causal direction.

Desiderata analysis. Following the convention of Section 3, we use $R(A \rightsquigarrow B)$ for the method-agnostic responsibility and ϕ_A^B exclusively for SHAP/Causal SHAP values; prose statements about desiderata refer to magnitudes $|R(A \rightsquigarrow B)|$. Here A denotes the candidate cause and B the target outcome (the generic Y of Section 3); for the PCI columns B^n, B^s are its necessity/sufficiency worlds, $B^\star$ its factual value, and $\bar{c} = |B^n - B^\star| - |B^s - B^\star|$ the absolute-difference impact of Example 21. Table 7 maps the desiderata from Section 4.3 to the values computed above.

Desideratum	Condition	Plain SHAP	Causal SHAP	PCI ($\mathbf{W}{=}\emptyset$)	PCI ($\mathbf{W}$=3rd)
D-XY	$\lvert R(X \rightsquigarrow Y)\rvert > 0$	0.250 ✓	0.250 ✓	0.249 ✓	0.186 ✓†
D-MY	$\lvert R(M \rightsquigarrow Y)\rvert > 0$	0.250 ✓	0.250 ✓	0.283 ✓	0.319 ✓
D-XM	$\lvert R(X \rightsquigarrow M)\rvert > 0$	0.306 ✓	0.375 ✓	0.253 ✓	0.284 ✓
D-YX	$\lvert R(Y \rightsquigarrow X)\rvert = 0$	0.139 ×	0.000 ✓	0.000 ✓	0.000 ✓
D-YM	$\lvert R(Y \rightsquigarrow M)\rvert = 0$	0.194 ×	0.125 ×	0.000 ✓	0.000 ✓
D-MX	$\lvert R(M \rightsquigarrow X)\rvert = 0$	0.219 ×	0.357 ×	0.000 ✓	0.000 ✓
D-MXY	$\lvert R(M \rightsquigarrow Y)\rvert > \lvert R(X \rightsquigarrow Y)\rvert$	0.25=0.25 ×	0.25=0.25 ×	0.283>0.249 ✓	0.319>0.186 ✓‡

Table 7: Desiderata satisfied (✓) and violated (×) by plain SHAP, Causal SHAP, and PCI on the signal-with-mediation example ($X \to M \to Y$), instance $X = M = Y = 1$. PCI uses $\bar{c} = \mathbb{E}[|B^n - B^\star|] - \mathbb{E}[|B^s - B^\star|]$ with Γ_s, Γ_w uniform over non-empty subsets and rejection on $\mathbf{C} \cap \mathbf{T} \neq \emptyset$. **W=3rd** sets the third variable (neither A nor B) as the witness when not in $\mathbf{C}$. † 0.186 weights three suspect-containing (C, T) configurations at $1/4$ each (the rejection-sampling filter keeps $2/3$ of $\Gamma_s \times \Gamma_w$ draws, including a fourth valid pair $(\{M\}, \emptyset)$ that does not contain X): one pins M as witness (contribution 0, necessity and sufficiency coincide) and two let M propagate freely (contributions 0.320 and 0.425). ‡ The two values use different witness sets ($\mathbf{W}{=}\{M\}$ for D-XY, $\mathbf{W}{=}\{X\}$ for D-MY); this column reports the amplified verdict under an asymmetric witness rule. The $\mathbf{W}{=}\emptyset$ column above is the corresponding single-game ranking and already passes D-MXY.

Positive results (D-XY, D-MY, D-XM). Both methods correctly assign positive attribution to every direct or upstream cause of the target. Causal prediction models are built from observational conditionals, which do preserve correlation from cause to effect, so these desiderata pose no challenge to either method.

Causal SHAP fixes D-YX, not D-YM. For target X, the interventional value $v_{\text{causal}}(\{Y\}) = 0.5$ equals the baseline because M is not downstream of Y: $P(M \mid do(Y{=}1)) = P(M)$, so fixing $Y = 1$ provides no interventional leverage over f_X, and Causal SHAP correctly yields $\phi_Y^X = 0$. For target M, however, $\phi_Y^M = 0.125 > 0$ persists. The prediction model $f_M(X, Y) = 0.5X + 0.5Y$ contains Y as an input because Y is observationally predictive of M; setting $Y = 1$ raises the model output even under the interventional regime. Causal SHAP replaces the conditioning distribution but cannot remove a feature that the prediction model itself uses.

D-MX worsens under Causal SHAP. Plain SHAP assigns $\phi_M^X = 0.219$; Causal SHAP assigns 0.357. Fixing D-YX reduces $v_{\text{causal}}(\{Y\})$ from 0.778 to 0.5, driving ϕ_Y^X to zero, but reallocates the freed attribution to M. Since the structural noise that actually drives X is unobserved, all attribution concentrates on the only visible predictor, M, even though M does not cause X.

D-MXY fails for both. Since $f_Y = M$, the singleton values $v(\{X\}) = \mathbb{E}[M \mid X{=}1] = 1$ and $v(\{M\}) = 1$ are identical, giving $\phi_M^Y = \phi_X^Y = 0.25$ for both methods. The Shapley formula cannot distinguish that X reaches Y only through M: when $X = 1$, the expected prediction is already 1 regardless of whether M is observed or not, so both features receive the same marginal contribution.

PCI: structural intervention separates the paths; witnesses sharpen the gap. Table 7 reports PCI in two configurations: without witnesses ($\mathbf{W} = \emptyset$) and with the third variable as witness.

Without witnesses, PCI satisfies all six single-variable desiderata. D-YX, D-YM, and D-MX are zero exactly: intervening on a variable that does not structurally affect the target leaves both the necessity and sufficiency worlds with identical distributions, so $ci = 0$ identically. D-XY, D-MY, and D-XM are positive. D-MXY also **passes** at $\mathrm{PCI}(M \to Y) \approx 0.283 > \mathrm{PCI}(X \to Y) \approx 0.249$, but only by a margin of 0.034. The asymmetry comes from the sufficiency world: at $C{=}\{X\}$, X's path to Y accumulates $\varepsilon_M + \varepsilon_Y$ in $|Y^s - Y^\star|$, while at $C{=}\{M\}$, M's path accumulates only ε_Y. The necessity expectations match across paths ($\mathrm{Var}(X) + \mathrm{Var}(\varepsilon_M) + \mathrm{Var}(\varepsilon_Y) = \mathrm{Var}(M) + \mathrm{Var}(\varepsilon_Y) = 0.45$), so the gap is carried entirely by the sufficiency variance asymmetry $\mathrm{Var}(\varepsilon_M)$. A small structural margin like this is fragile: under any perturbation of the noise variances or the contrast function the $\mathbf{W} = \emptyset$ ordering could flip.

Admitting the third variable as a witness amplifies the gap by roughly fourfold. $\mathrm{PCI}(X \to Y \mid \mathbf{W}{=}\{M\}) \approx 0.186$ weights three suspect-containing (C, T) configurations at $1/4$ each: in two cases M propagates freely (contributing 0.320 and 0.425) and in one case M is pinned as witness in both worlds, equalizing them and contributing 0. $\mathrm{PCI}(M \to Y \mid \mathbf{W}{=}\{X\}) \approx 0.319$ weights three cases in which M takes an alternative value; since X does not appear in Y's structural equation, pinning X has no effect on Y^n or Y^s, and all three cases contribute the same 0.425. The gap widens from 0.034 at $\mathbf{W} = \emptyset$ to $0.319 - 0.186 = 0.133$ at $\mathbf{W} = \{\text{third}\}$: the ordering is now *robust* rather than marginal. The comparison relies on different witness sets for the two cells, so it remains illustrative rather than a single-game ranking.

Instance 2 (baseline). We now return to **Instance 2**: every variable sits at its mean, $X^\star = M^\star = Y^\star = 0.5$, and the model predicts the baseline at every target: $f_Y(0.5, 0.5) = 0.5$, $f_M(0.5, 0.5) = 0.5$, $f_X(0.5, 0.5) = 0.5$. At this instance, SHAP's attribution budget $f(x^\star) - \mathbb{E}[f(X)]$ vanishes, while PCI's machinery is unaffected.

	Plain / Causal SHAP		PCI ($\mathbf{W} = \emptyset$)		PCI ($\mathbf{W} = \{\text{third}\}$)	
desideratum	value	status	value	status	value	status
D-XY	0.000	×	0.154	✓	0.115	✓
D-MY	0.000	×	0.189	✓	0.212	✓
D-XM	0.000	×	0.146	✓	0.165	✓
D-YX	0.000	✓	0.000	✓	0.000	✓
D-YM	0.000	✓	0.000	✓	0.000	✓
D-MX	0.000	✓	0.000	✓	0.000	✓
D-MXY	$0.000 = 0.000$ ×		$0.189 > 0.154$ ✓		$0.212 > 0.115$ ✓	

Table 8: Desiderata at the baseline instance $X^\star = M^\star = Y^\star = 0.5$. SHAP collapses to zero on every cell, correctly assigning zero to the non-causal triples by accident, but with no signal to offer on the causal ones. PCI reports the same structural verdicts as at $X^\star = M^\star = Y^\star = 1$.

The pattern in Table 8 reflects the object each method is decomposing. SHAP attributes the quantity $f(x^\star) - \mathbb{E}[f(X)]$ across features. At the baseline this gap is zero on every target, so the budget being distributed is zero, and every cell collapses to zero, uniformly, regardless of how the realized outcome was actually produced. The non-causal triples (D-YX, D-YM, D-MX) are satisfied only by this collapse: SHAP is not isolating the absence of a causal path; it is reporting that there is nothing to attribute. The same collapse turns D-XY, D-MY, and D-XM into violations.

PCI never invokes a population expectation. With $\mathbf{S}$ and $\mathbf{W}$ defined relative to a fixed factual instance, the question is whether varying the suspect would move the realized outcome $y^\star$ away from itself, not whether $f(x^\star)$ deviates from $\mathbb{E}[f(X)]$. The structural mechanism is unchanged at the baseline, so the three causal pairs continue to register positive values; the three non-causal pairs continue to register zero exactly, because intervening on a variable that does not structurally affect the target leaves both worlds with identical distributions. The D-MXY verdict is preserved in both witness configurations, with margin 0.035 at $\mathbf{W} = \emptyset$ (vs. 0.034 at the $X^\star = M^\star = Y^\star = 1$ instance) and 0.097 at $\mathbf{W} = \{\text{third}\}$ (vs. 0.133). The gap is a property of the PSCM's noise variances, not of where the factual sits.